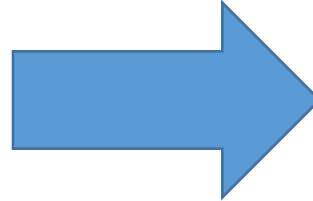


# Electrical Machine- II (BT EE403A)



हरियाणा केंद्रीय विश्वविद्यालय  
**CENTRAL UNIVERSITY OF HARYANA**  
NAAC ACCREDITED 'A' GRADE UNIVERSITY

## Unit-II



### Syllabus/Topics Covered

Three Phase Induction Motors: Types, Construction and principle of operation , 3phase Induction Motor, general Diagram, equivalent circuit, power and torque relations, condition for maximum torque, circle diagram, performance characteristics, effect of rotor resistance on speed torque characteristics, stable & unstable region of operation, operation with unbalanced supply voltage. Starting: Starting of 3 phase induction motors, high starting torque motors, speed control, rheostatic method, pole changing method cascade control of speed, Double age induction motor, Cogging and Crawling of induction motor, induction generator

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# Three phase Induction Motor

The most common type of AC motor being used throughout the world today is the "Induction Motor". Applications of three-phase induction motors of size varying from half a kilowatt to thousands of kilowatts are numerous. They are found everywhere from a small workshop to a large manufacturing industry.

The advantages of three-phase AC induction motor are listed below:

- Simple design
- Rugged construction
- Reliable operation
- Low initial cost
- Easy operation and simple maintenance
- Simple control gear for starting and speed control
- High efficiency.

### 3.2 Types and Construction of Three Phase Induction Motor

Three phase induction motors are constructed into two major types:

1. Squirrel cage Induction Motors
2. Slip ring Induction Motors

#### 3.2.1 *Squirrel cage Induction Motors*

##### *(a) Stator Construction*

The induction motor stator resembles the stator of a revolving field, three phase alternator. The stator or the stationary part consists of three phase winding held in place in the slots of a laminated steel core which is enclosed and supported by a cast iron or a steel frame as shown in Fig: 3.1(a).

The phase windings are placed 120 electrical degrees apart and may be connected in either star or delta externally, for which six leads are brought out to a terminal box mounted on the frame of the motor. When the stator is energized from a three phase voltage it will produce a rotating magnetic field in the stator core.

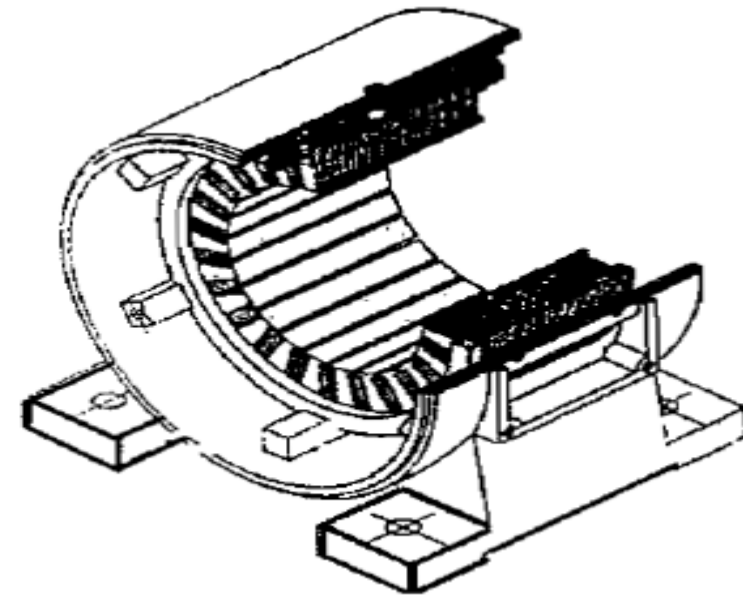


Fig:3.1 (a)

### *(b) Rotor Construction*

The rotor of the squirrel cage motor shown in Fig: 3.1(b) contains no windings. Instead it is a cylindrical core constructed of steel laminations with conductor bars mounted parallel to the shaft and embedded near the surface of the rotor core.

These conductor bars are short circuited by an end rings at both end of the rotor core. In large machines, these conductor bars and the end rings are made up of copper with the bars brazed or welded to the end rings shown in Fig: 3.1(b). In small machines the conductor bars and end rings are sometimes made of aluminium with the bars and rings cast in as part of the rotor core. Actually the entire construction (bars and end-rings) resembles a squirrel cage, from which the name is derived.

The rotor or rotating part is not connected electrically to the power supply but has voltage induced in it by transformer action from the stator. For this reason, the stator is sometimes called the primary and the rotor is referred to as the secondary of the motor since the motor operates on the principle of induction and as the construction of the rotor with the bars and end rings resembles a squirrel cage, the squirrel cage induction motor is used.

The rotor bars are not insulated from the rotor core because they are made of metals having less resistance than the core. The induced current will flow mainly in them. Also the rotor bars are usually not quite parallel to the rotor shaft but are mounted in a slightly skewed position. This feature tends to produce a more uniform rotor field and torque. Also it helps to reduce some of the internal magnetic noise when the motor is running.

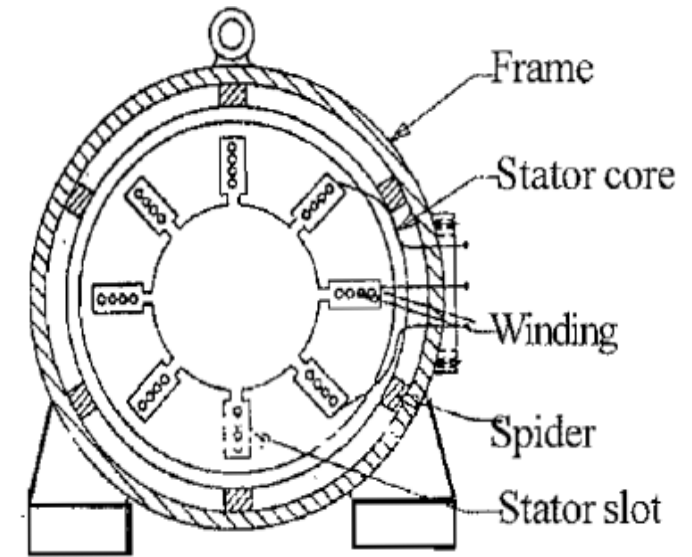


Fig. 3.1(b)

### (c) End Shields

The function of the two end shields is to support the rotor shaft. They are fitted with bearings and attached to the stator frame with the help of studs or bolts attention.

### 3.2.2 Slip ring Induction Motors

#### *(a) Stator Construction*

The construction of the slip ring induction motor is exactly similar to the construction of squirrel cage induction motor. There is no difference between squirrel cage and slip ring motors.

#### *(b) Rotor Construction*

The rotor of the slip ring induction motor is also cylindrical or constructed of lamination.

Squirrel cage motors have a rotor with short circuited bars whereas slip ring motors have wound rotors having "three windings" each connected in star.

The winding is made of copper wire. The terminals of the rotor windings of the slip ring motors are brought out through slip rings which are in contact with stationary brushes as shown in Fig: 3.2.

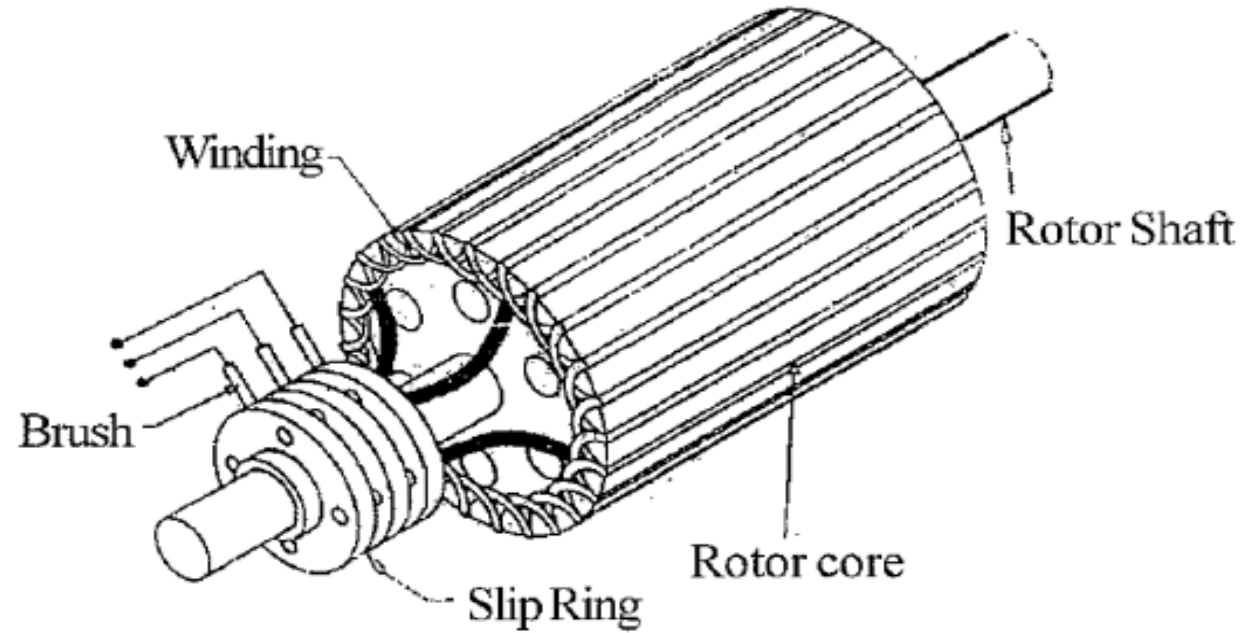


Fig: 3.2

THE ADVANTAGES OF THE SLIPRING MOTOR ARE

- It has susceptibility to speed control by regulating rotor resistance.
- High starting torque of 200 to 250% of full load value.
- Low starting current of the order of 250 to 350% of the full load current.

Hence slip ring motors are used where one or more of the above requirements are to be met.

### 3.2.3 Comparison of Squirrel Cage and Slip Ring Motor

| Sl.No. | Property                  | <i>Squirrel cage motor</i>                                                                                                                              | <i>Slip ring motor</i>                                                                                                                                                                    |
|--------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.     | <b>Rotor Construction</b> | <i>Bars are used in rotor. Squirrel cage motor is very simple, rugged and long lasting. No slip rings and brushes</i>                                   | <i>Winding wire is to be used.<br/><br/>Wound rotor required attention.<br/><br/>Slip ring and brushes are needed also need frequent maintenance.</i>                                     |
| 2.     | <b>Starting</b>           | <i>Can be started by D.O.L., star-delta, auto transformer starters</i>                                                                                  | <i>Rotor resistance starter is required.</i>                                                                                                                                              |
| 3.     | <b>Starting torque</b>    | <i>Low</i>                                                                                                                                              | <i>Very high</i>                                                                                                                                                                          |
| 4.     | <b>Starting Current</b>   | <i>High</i>                                                                                                                                             | <i>Low</i>                                                                                                                                                                                |
| 5.     | <b>Speed variation</b>    | <i>Not easy, but could be varied in large steps by pole changing or through smaller incremental steps through thyristors or by frequency variation.</i> | <i>Easy to vary speed.<br/><br/>Speed change is possible by inserting rotor resistance using thyristors or by using frequency variation injecting emf in the rotor circuit cascading.</i> |
| 6.     | <b>Maintenance</b>        | <i>Almost ZERO maintenance</i>                                                                                                                          | <i>Requires frequent maintenance</i>                                                                                                                                                      |
| 7.     | <b>Cost</b>               | <i>Low</i>                                                                                                                                              | <i>High</i>                                                                                                                                                                               |

### 3.3 Principle of Operation

The operation of a 3-phase induction motor is based upon the application of Faraday Law and the Lorentz force on a conductor. The behaviour can readily be understood by means of the following example.

Consider a series of conductors of length  $l$ , whose extremities are short-circuited by two bars A and B (Fig.3.3 a). A permanent magnet placed above this conducting ladder, moves rapidly to the right at a speed  $v$ , so that its magnetic field  $B$  sweeps across the conductors. The following sequence of events then takes place:

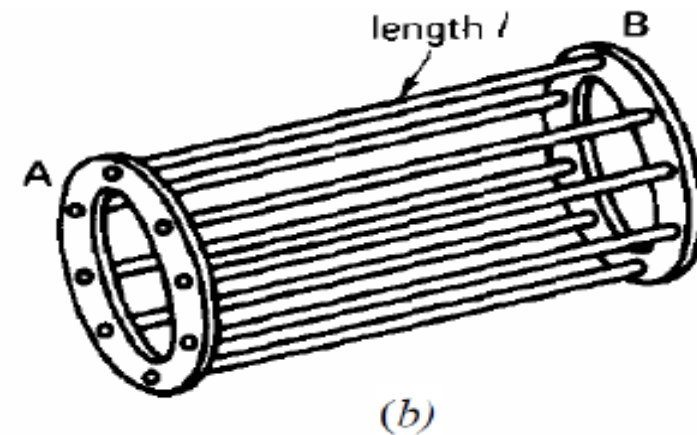
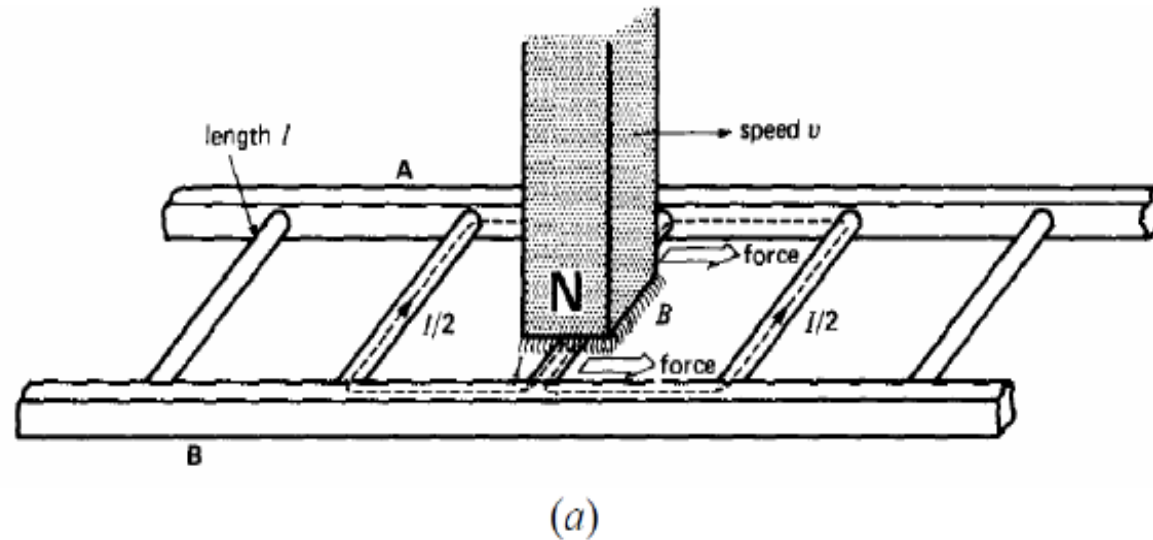


Fig. 3.3

1. A voltage  $E = Blv$  is induced in each conductor while it is being cut by the flux (Faraday law).
2. The induced voltage immediately produces a current  $I$ , which flows down the conductor underneath the pole face, through the end-bars, and back through the other conductors.
3. Because the current carrying conductor lies in the magnetic field of the permanent magnet, it experiences a mechanical force (Lorentz force).
4. The force always acts in a direction to drag the conductor along with the magnetic field. If the conducting ladder is free to move, it will accelerate toward the right. However, as it picks up speed, the conductors will be cut less rapidly by the moving magnet, with the result that the induced voltage  $E$  and the current  $I$  will diminish. Consequently, the force acting on the conductors will also decrease. If the ladder were to move at the same speed as the magnetic field, the induced voltage  $E$ , the current  $I$ , and the force dragging the ladder along would all become zero.

In an induction motor the ladder is closed upon itself to form a squirrel-cage (Fig.3.3b) and the moving magnet is replaced by a rotating field. The field is produced by the 3-phase currents that flow in the stator windings.

### 3.4 Rotating Magnetic Field and Induced Voltages

Consider a simple stator having 6 salient poles, each of which carries a coil having 5 turns (Fig.3.4). Coils that are diametrically opposite are connected in series by means of three jumpers that respectively connect terminals a-a, b-b, and c-c. This creates three identical sets of windings AN, BN, CN, which are mechanically spaced at 120 degrees to each other. The two coils in each winding produce magneto motive forces that act in the same direction.

The three sets of windings are connected in wye, thus forming a common neutral N. Owing to the perfectly symmetrical arrangement, the line to neutral impedances are identical. In other words, as regards terminals A, B, C, the windings constitute a balanced 3-phase system.

For a two-pole machine, rotating in the air gap, the magnetic field (i.e., flux density) being sinusoidally distributed with the peak along the centre of the magnetic poles. The result is illustrated in Fig.3.5. The rotating field will induce voltages in the phase coils aa', bb', and cc'. Expressions for the induced voltages can be obtained by using Faraday laws of induction.

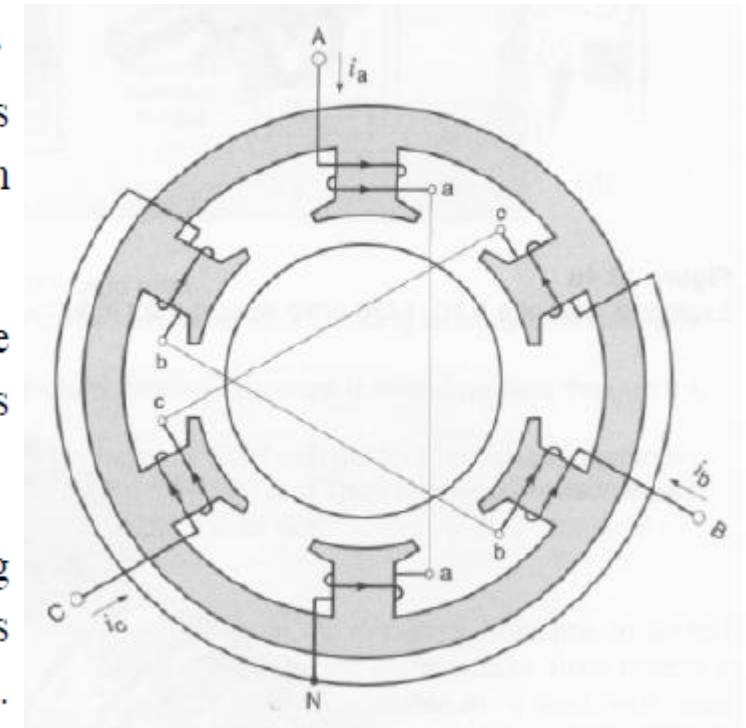


Fig. 3.4

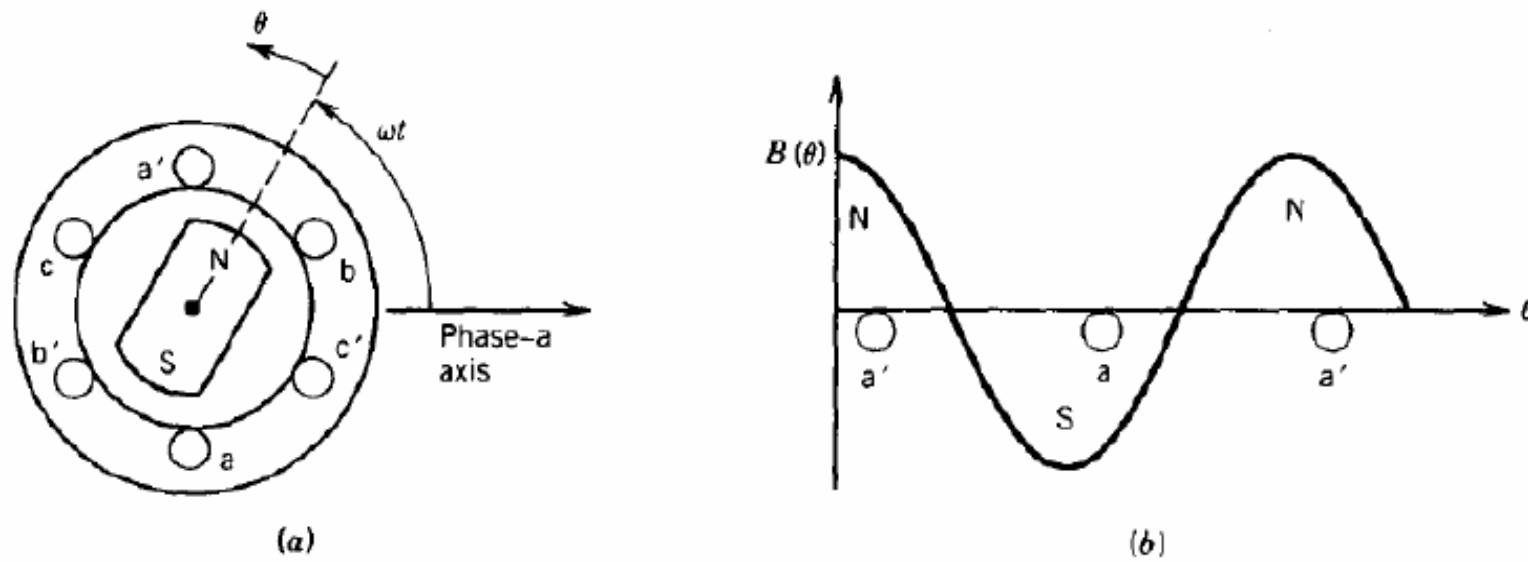


Fig: 3.5 Air gap flux density distribution.

The flux density distribution in the air gap can be expressed as:

$$B(\theta) = B_{\max} \cos \theta$$

The air gap flux per pole,  $\phi_P$ , is:

$$\phi_P = \int_{-\pi/2}^{\pi/2} B(\theta) l r d\theta = 2B_{\max} l r$$

Where,

$l$  is the axial length of the stator.

$r$  is the radius of the stator at the air gap.

Let us consider that the phase coils are full-pitch coils of  $N$  turns (the coil sides of each phase are 180 electrical degrees apart as shown in Fig.3.5). It is obvious that as the rotating field moves (or the magnetic poles rotate) the flux linkage of a coil will vary. The flux linkage for coil  $aa'$  will be maximum.

( $= N \phi_p$  at  $\omega t = 0^\circ$ ) (Fig.3.5a) and zero at  $\omega t = 90^\circ$ . The flux linkage  $\lambda_a(\omega t)$  will vary as the cosine of the angle  $\omega t$ .

Hence,

$$\lambda_a(\omega t) = N \phi_p \cos \omega t$$

Therefore, the voltage induced in phase coil **aa'** is obtained from *Faraday law* as:

$$e_a = - \frac{d\lambda_a(\omega t)}{dt} = \omega N \phi_p \sin \omega t = E_{\max} \sin \omega t$$

The voltages induced in the other phase coils are also sinusoidal, but phase-shifted from each other by 120 electrical degrees. Thus,

$$e_b = E_{\max} \sin(\omega t - 120)$$

$$e_c = E_{\max} \sin(\omega t + 120).$$

the *rms* value of the induced voltage is:

$$E_{rms} = \frac{\omega N \phi_p}{\sqrt{2}} = \frac{2\pi f}{\sqrt{2}} N \phi_p = 4.44 f N \phi_p$$

Where *f* is the frequency in hertz. Above equation has the same form as that for the induced voltage in transformers. However,  $\phi_p$  represents the flux per pole of the machine.

The above equation also shows the rms voltage per phase. The *N* is the total number of series turns per phase with the turns forming a concentrated full-pitch winding. In an actual AC machine each phase winding is distributed in a number of slots for better use of the iron and copper and to improve the waveform. For such a distributed winding, the EMF induced in various coils placed in different slots are not in time phase, and therefore the phasor sum of the EMF is less than their numerical sum when they are connected in series for the phase winding. A reduction factor *K<sub>w</sub>*, called the winding factor, must therefore be applied. For most three-phase machine windings *K<sub>w</sub>* is about 0.85 to 0.95.

Therefore, for a distributed phase winding, the rms voltage per phase is

$$E_{rms} = 4.44 f N_{ph} \phi_p K_w$$

Where *N<sub>ph</sub>* is the number of turns in series per phase.

### 3.5 Alternate Analysis for Rotating Magnetic Field

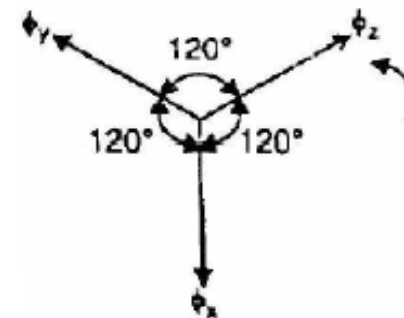
When a 3-phase winding is energized from a 3-phase supply, a rotating magnetic field is produced. This field is such that its poles do not remain in a fixed position on the stator but go on shifting their positions around the stator. For this reason, it is called a rotating field. It can be shown that magnitude of this rotating field is constant and is equal to  $1.5 m$  where  $m$  is the maximum flux due to any phase.

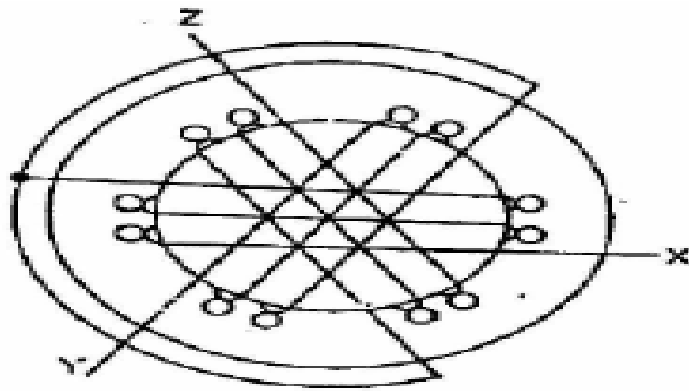
To see how rotating field is produced, consider a 2-pole, 3-phase winding as shown in Fig. 3.6 (i). The three phases X, Y and Z are energized from a 3-phase source and currents in these phases are indicated as  $I_x$ ,  $I_y$  and  $I_z$  [See Fig. 3.6 (ii)]. Referring to Fig. 3.6 (ii), the fluxes produced by these currents are given by:

$$\phi_x = \phi_m \sin \omega t$$

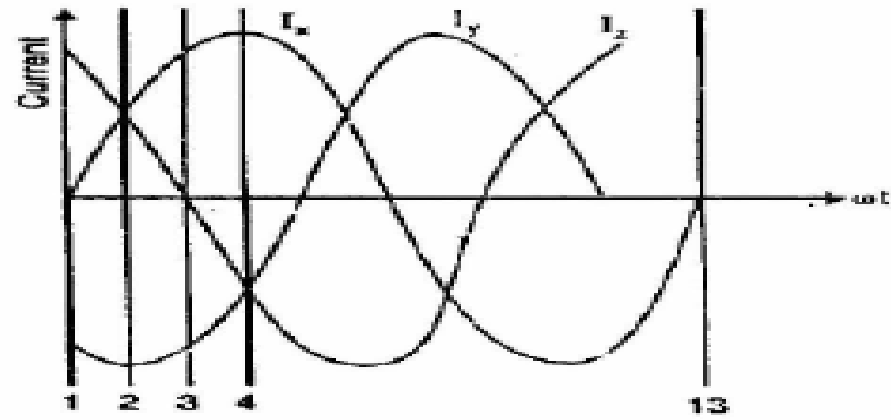
$$\phi_y = \phi_m \sin (\omega t - 120^\circ)$$

$$\phi_z = \phi_m \sin (\omega t - 240^\circ)$$

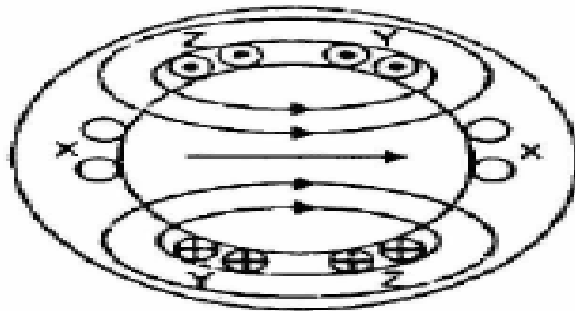




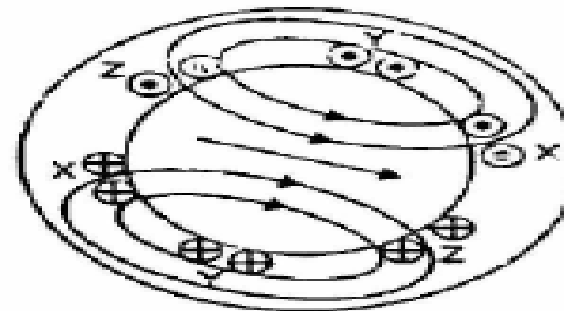
(i)



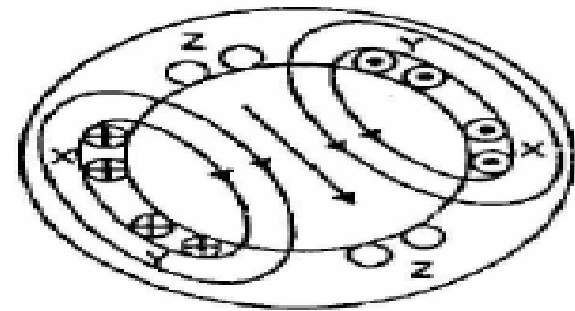
(ii)



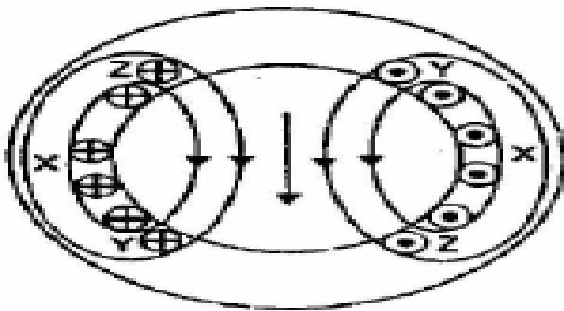
(1)



(2)



(3)



(4)

(iii)

(iii)

Fig: 3.6

Here  $\phi_m$  is the maximum flux due to any phase. Above figure shows the phasor diagram of the three fluxes. We shall now prove that this 3-phase supply produces a rotating field of constant magnitude equal to  $1.5 \phi_m$ .

At instant 1 [See Fig. 3.6 (ii) and Fig. 3.6 (iii)], the current in phase X is zero and currents in phases Y and Z are equal and opposite. The currents are flowing outward in the top conductors and inward in the bottom conductors. This establishes a resultant flux towards right. The magnitude of the resultant flux is constant and is equal to  $1.5 \phi_m$  as proved under:

At instant 1,  $\omega t = 0^\circ$ . Therefore, the three fluxes are given by;

$$\phi_x = 0; \quad \phi_y = \phi_m \sin(-120^\circ) = -\frac{\sqrt{3}}{2} \phi_m;$$

$$\phi_z = \phi_m \sin(-240^\circ) = \frac{\sqrt{3}}{2} \phi_m$$

The phasor sum of  $-\phi_y$  and  $\phi_z$  is the resultant flux  $\phi_r$

So,

$$\text{Resultant flux, } \phi_r = 2 \times \frac{\sqrt{3}}{2} \phi_m \cos \frac{60^\circ}{2} = 2 \times \frac{\sqrt{3}}{2} \phi_m \times \frac{\sqrt{3}}{2} = 1.5 \phi_m$$

At instant 2 [Fig: 3.7 (ii)], the current is maximum (negative) in  $\phi_y$  phase Y and 0.5 maximum (positive) in phases X and Z. The magnitude of resultant flux is  $1.5 \phi_m$  as proved under:

At instant 2,  $\omega t = 30^\circ$ . Therefore, the three fluxes are given by;

$$\phi_x = \phi_m \sin 30^\circ = \frac{\phi_m}{2}$$

$$\phi_y = \phi_m \sin (-90^\circ) = -\phi_m$$

$$\phi_z = \phi_m \sin (-210^\circ) = \frac{\phi_m}{2}$$

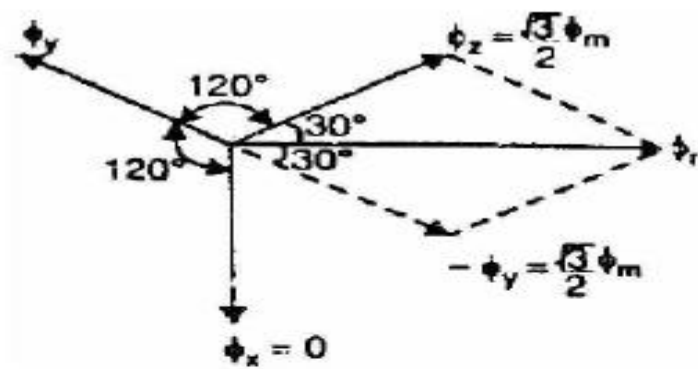
The phasor sum of  $\phi_x$ ,  $-\phi_y$  and  $\phi_z$  is the resultant flux  $\phi_r$

$$\text{Phasor sum of } \phi_x \text{ and } \phi_z, \phi'_r = 2 \times \frac{\phi_m}{2} \cos \frac{120^\circ}{2} = \frac{\phi_m}{2}$$

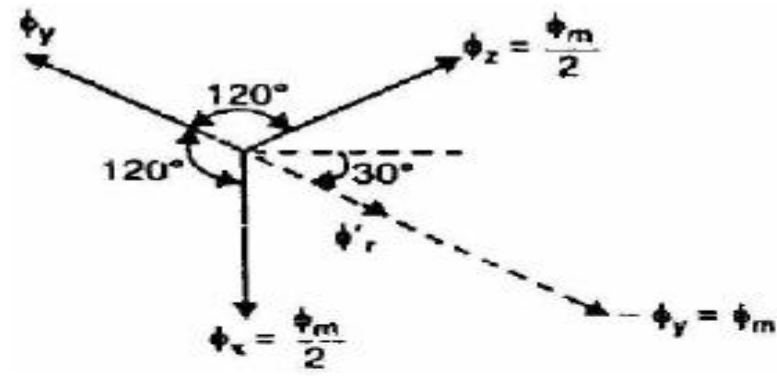
$$\text{Phasor sum of } \phi'_r \text{ and } -\phi_y, \phi_r = \frac{\phi_m}{2} + \phi_m = 1.5 \phi_m$$

Note that resultant flux is displaced  $30^\circ$  clockwise from position 1.

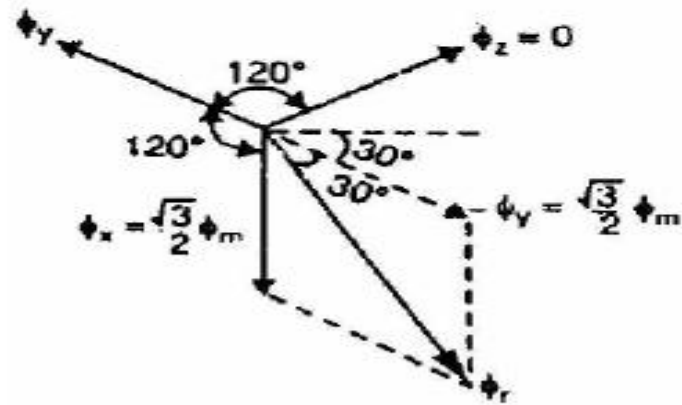
At instant 3 [Fig: 3.7 (iii)], current in phase Z is zero and the currents in phases X and Y are equal and opposite (currents in phases X and Y are  $0.866 \times \text{max. value}$ ). The magnitude of resultant flux is  $1.5 \phi_m$  as proved under:



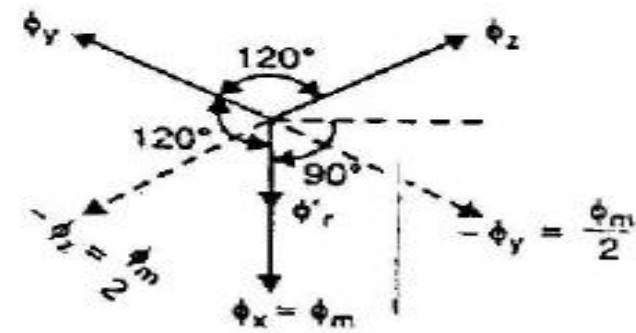
(i)



(ii)



(iii)



(iv)

Fig: 3.7

At instant 3,  $\omega t = 60^\circ$ . Therefore, the three fluxes are given by;

$$\phi_x = \phi_m \sin 60^\circ = \frac{\sqrt{3}}{2} \phi_m;$$

$$\phi_y = \phi_m \sin(-60^\circ) = -\frac{\sqrt{3}}{2} \phi_m;$$

$$\phi_z = \phi_m \sin(-180^\circ) = 0$$

The resultant flux  $\phi_r$  is the phasor sum of  $\phi_x$  and  $-\phi_y$  ( $\because \phi_z = 0$ ).

$$\phi_r = 2 \times \frac{\sqrt{3}}{2} \phi_m \cos \frac{60^\circ}{2} = 1.5 \phi_m$$

Note that resultant flux is displaced  $60^\circ$  clockwise from position 1.

At instant 4 [Fig: 3.7 (iv)], the current in phase X is maximum (positive) and the currents in phases V and Z are equal and negative (currents in phases V and Z are  $0.5 \times \text{max. value}$ ). This establishes a resultant flux downward as shown under:

At instant 4,  $\omega t = 90^\circ$ . Therefore, the three fluxes are given by;

$$\phi_x = \phi_m \sin 90^\circ = \phi_m$$

$$\phi_y = \phi_m \sin (-30^\circ) = -\frac{\phi_m}{2}$$

$$\phi_z = \phi_m \sin (-150^\circ) = -\frac{\phi_m}{2}$$

The phasor sum of  $\phi_x$ ,  $-\phi_y$  and  $-\phi_z$  is the resultant flux  $\phi_r$

$$\text{Phasor sum of } -\phi_z \text{ and } -\phi_y, \phi'_r = 2 \times \frac{\phi_m}{2} \cos \frac{120^\circ}{2} = \frac{\phi_m}{2}$$

$$\text{Phasor sum of } \phi'_r \text{ and } \phi_x, \phi_r = \frac{\phi_m}{2} + \phi_m = 1.5 \phi_m$$

Note that the resultant flux is downward i.e., it is displaced  $90^\circ$  clockwise from position 1.

It follows from the above discussion that a 3-phase supply produces a rotating field of constant value ( $= 1.5 \phi_m$ , where  $\phi_m$  is the maximum flux due to any phase).

### 3.5.1 Speed of rotating magnetic field

The speed at which the rotating magnetic field revolves is called the synchronous speed ( $N_s$ ). Referring to Fig. 3.6 (ii), the time instant 4 represents the completion of one-quarter cycle of alternating current  $I_x$  from the time instant 1. During this one quarter cycle, the field has rotated through  $90^\circ$ . At a time instant represented by 13 [Fig. 3.6 (ii)] or one complete cycle of current  $I_x$  from the origin, the field has completed one revolution. Therefore, for a 2-pole stator winding, the field makes one revolution in one cycle of current. In a 4-pole stator winding, it can be shown that the rotating field makes one revolution in two cycles of current. In general, for  $P$  poles, the rotating field makes one revolution in  $P/2$  cycles of current.

$$\therefore \quad \text{Cycles of current} = \frac{P}{2} \times \text{revolutions of field}$$

$$\text{or} \quad \text{Cycles of current per second} = \frac{P}{2} \times \text{revolutions of field per second}$$

Since revolutions per second is equal to the revolutions per minute ( $N_s$ ) divided by 60 and the number of cycles per second is the frequency  $f$ ,

$$\therefore \quad f = \frac{P}{2} \times \frac{N_s}{60} = \frac{N_s P}{120}$$

$$\text{or} \quad N_s = \frac{120 f}{P}$$

The speed of the rotating magnetic field is the same as the speed of the alternator that is supplying power to the motor if the two have the same number of poles. Hence the magnetic flux is said to rotate at synchronous speed.

### 3.5.2 Direction of rotating magnetic field

The phase sequence of the three-phase voltage applied to the stator winding in Fig. 3.6 (ii) is X-Y-Z. If this sequence is changed to X-Z-Y, it is observed that direction of rotation of the field is reversed i.e., the field rotates counter clockwise rather than clockwise. However, the number of poles and the speed at which the magnetic field rotates remain unchanged. Thus it is necessary only to change the phase sequence in order to change the direction of rotation of

the magnetic field. For a three-phase supply, this can be done by interchanging any two of the three lines. As we shall see, the rotor in a 3-phase induction motor runs in the same direction as the rotating magnetic field. Therefore, the direction of rotation of a 3-phase induction motor can be reversed by interchanging any two of the three motor supply lines.

### 3.5.3 Slip

We have seen above that rotor rapidly accelerates in the direction of rotating field. In practice, the rotor can never reach the speed of stator flux. If it did, there would be no relative speed between the stator field and rotor conductors, no induced rotor currents and, therefore, no torque to drive the rotor. The friction and windage would immediately cause the rotor to slow down. Hence, the rotor speed ( $N$ ) is always less than the stator field speed ( $N_s$ ). This difference in speed depends upon load on the motor. The difference between the synchronous speed  $N_s$  of the rotating stator field and the actual rotor speed  $N$  is called slip. It is usually expressed as a percentage of synchronous speed i.e.

$$\% \text{ age slip, } s = \frac{N_s - N}{N_s} \times 100$$

- (i) The quantity  $N_s - N$  is sometimes called slip speed.
- (ii) When the rotor is stationary (i.e.,  $N = 0$ ), slip,  $s = 1$  or 100 %.
- (iii) In an induction motor, the change in slip from no-load to full-load is hardly 0.1% to 3% so that it is essentially a constant-speed motor.

### 3.5.4 Rotor Current Frequency

The frequency of a voltage or current induced due to the relative speed between a revolving and a magnetic field is given by the general formula;

$$\text{Frequency} = \frac{NP}{120}$$

where  $N$  = Relative speed between magnetic field and the winding  
 $P$  = Number of poles

For a rotor speed  $N$ , the relative speed between the rotating flux and the rotor is  $N_s - N$ . Consequently, the rotor current frequency  $f'$  is given by;

$$\begin{aligned} f' &= \frac{(N_s - N)P}{120} \\ &= \frac{s N_s P}{120} & \left( \because s = \frac{N_s - N}{N_s} \right) \\ &= sf & \left( \because f = \frac{N_s P}{120} \right) \end{aligned}$$

i.e., Rotor current frequency = Fractional slip x Supply frequency

- (i) When the rotor is at standstill or stationary (i.e.,  $s = 1$ ), the frequency of rotor current is the same as that of supply frequency ( $f' = sf = 1 \times f = f$ ).
- (ii) As the rotor picks up speed, the relative speed between the rotating flux and the rotor decreases. Consequently, the slip  $s$  and hence rotor current frequency decreases.

### 3.6 Phasor Diagram of Three Phase Induction Motor

In a 3-phase induction motor, the stator winding is connected to 3-phase supply and the rotor winding is short-circuited. The energy is transferred magnetically from the stator winding to the short-circuited, rotor winding. Therefore, an induction motor may be considered to be a transformer with a rotating secondary (short-circuited). The stator winding corresponds to transformer primary and the rotor winding corresponds to transformer secondary. In view of the similarity of the flux and voltage conditions to those in a transformer, one can expect that the equivalent circuit of an induction motor will be similar to that of a transformer. Fig. 3.8 shows the equivalent circuit per phase for an induction motor. Let discuss the stator and rotor circuits separately.

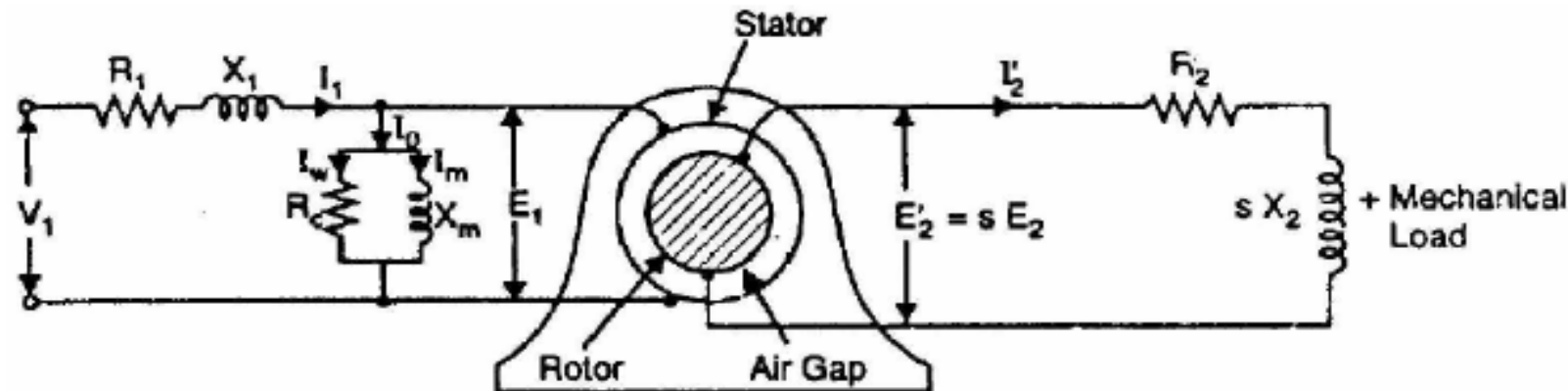


Fig: 3.8

**Stator circuit.** In the stator, the events are very similar to those in the transformer primary. The applied voltage per phase to the stator is  $V_1$  and  $R_1$  and  $X_1$  are the stator resistance and leakage reactance per phase respectively. The applied voltage  $V_1$  produces a magnetic flux which links the stator winding (i.e., primary) as well as the rotor winding (i.e., secondary). As a result, self-induced e.m.f.  $E_1$  is induced in the stator winding and mutually induced e.m.f.

$E'_2 (= s E_2 = s K E_1$  where  $K$  is transformation ratio) is induced in the rotor winding. The flow of stator current  $I_1$  causes voltage drops in  $R_1$  and  $X_1$ .

$$\therefore V_1 = -E_1 + I_1 (R_1 + j X_1) \text{ ...phasor sum}$$

When the motor is at no-load, the stator winding draws a current  $I_0$ . It has two components viz., (i) which supplies the no-load motor losses and (ii) magnetizing component  $I_m$  which sets up magnetic flux in the core and the air gap. The parallel combination of  $R_c$  and  $X_m$ , therefore, represents the no-load motor losses and the production of magnetic flux respectively.

$$\therefore I_0 = I_w + I_m$$

**Rotor circuit.** Here  $R_2$  and  $X_2$  represent the rotor resistance and standstill rotor reactance per phase respectively. At any slip  $s$ , the rotor reactance will be  $sX_2$ . The induced voltage/phase in the rotor is  $E'_2 = s E_2 = s K E_1$ . Since the rotor winding is short-circuited, the whole of e.m.f.  $E'_2$  is used up in circulating the rotor current  $I'_2$ .

$$\therefore E'_2 = I'_2 (R_2 + j s X_2)$$

The rotor current  $I'_2$  is reflected as  $I''_2 (= K I'_2)$  in the stator. The phasor sum of  $I''_2$  and  $I_0$  gives the stator current  $I_1$ .

It is important to note that input to the primary and output from the secondary of a transformer are electrical. However, in an induction motor, the inputs to the stator and rotor are electrical but the output from the rotor is mechanical. To facilitate calculations, it is desirable and necessary to replace the mechanical load by an equivalent electrical load. We then have the transformer equivalent circuit of the induction motor.

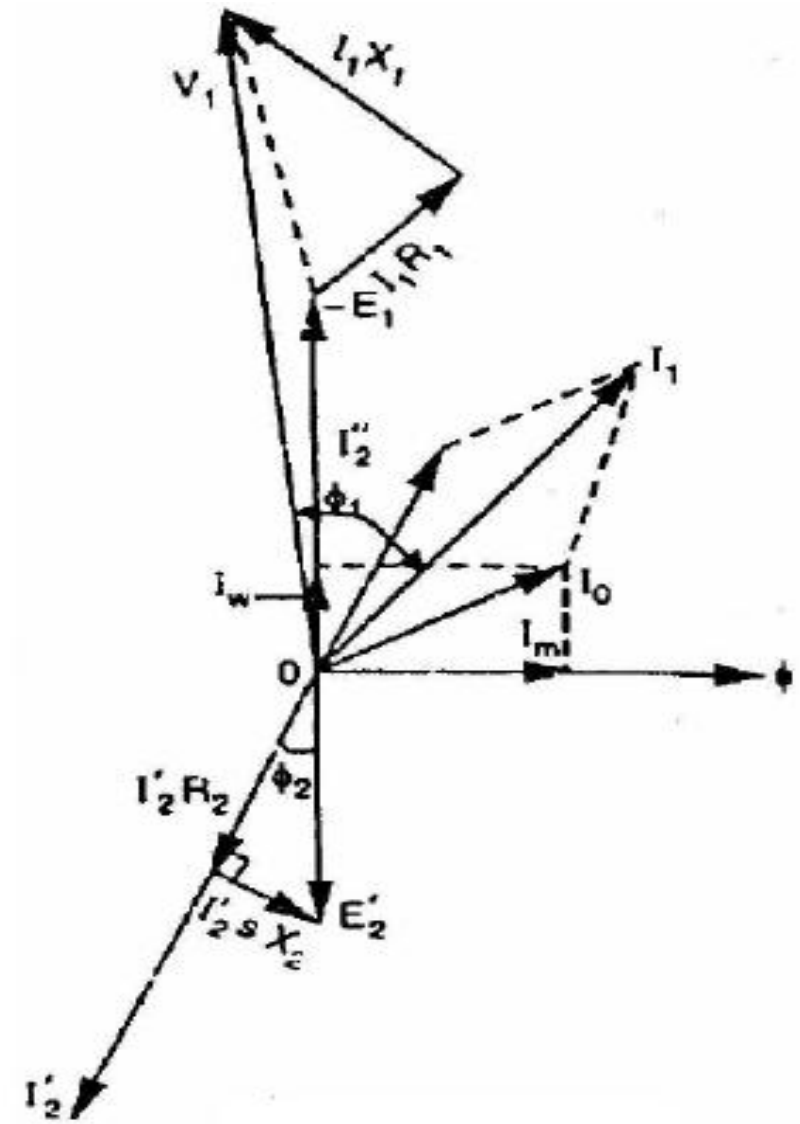


Fig: 3.9

It may be noted that even though the frequencies of stator and rotor currents are different, yet the magnetic fields due to them rotate at synchronous speed  $N_s$ . The stator currents produce a magnetic flux which rotates at a speed  $N_s$ . At slip  $s$ , the speed of rotation of the rotor field relative to the rotor surface in the direction of rotation of the rotor is

$$= \frac{120 f'}{P} = \frac{120 s f}{P} = s N_s$$

But the rotor is revolving at a speed of  $N$  relative to the stator core. Therefore, the speed of rotor field relative to stator core

$$= sN_s + N = (N_s - N) + N = N_s$$

Thus no matter what the value of slip  $s$ , the stator and rotor magnetic fields are synchronous with each other when seen by an observer stationed in space. Consequently, the 3-phase induction motor can be regarded as being equivalent to a transformer having an air-gap separating the iron portions of the magnetic circuit carrying the primary and secondary windings. Fig. 3.9 shows the phasor diagram of induction motor.

### 3.7 Equivalent Circuit of Three Phase Induction Motor

Fig. 3.10 (i) shows the equivalent circuit per phase of the rotor at slip  $s$ . The rotor phase current is given by;

$$I'_2 = \frac{s E_2}{\sqrt{R_2^2 + (s X_2)^2}}$$

Mathematically, this value is unaltered by writing it as:

$$I'_2 = \frac{E_2}{\sqrt{(R_2 / s)^2 + (X_2)^2}}$$

As shown in Fig. 3.10 (ii), we now have a rotor circuit that has a fixed reactance  $X_2$  connected in series with a variable resistance  $R_2/s$  and supplied with constant voltage  $E_2$ . Note that Fig. 3.10 (ii) transfers the variable to the resistance without altering power or power factor conditions.

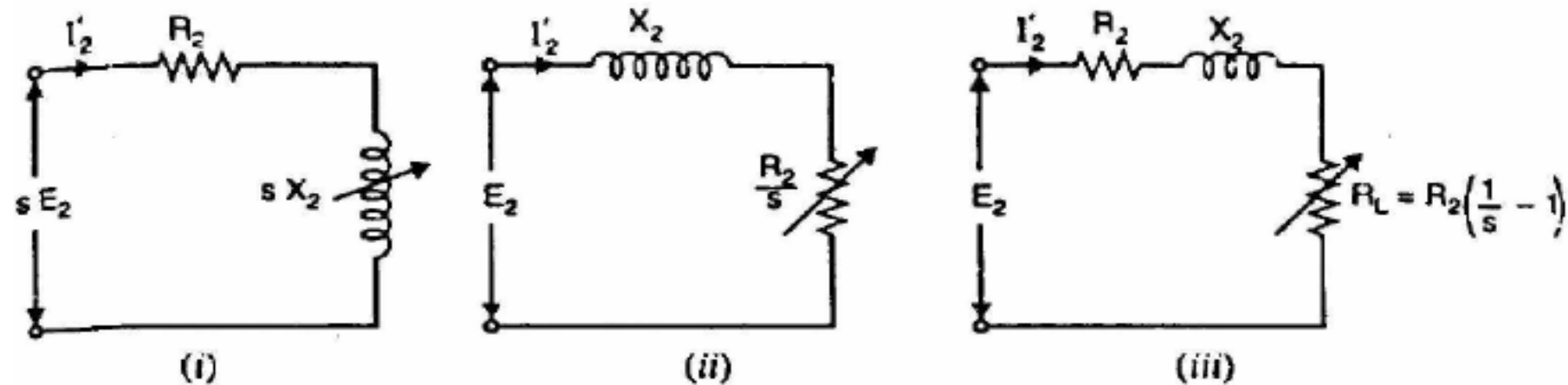


Fig: 3.10

The quantity  $R_2/s$  is greater than  $R_2$  since  $s$  is a fraction. Therefore,  $R_2/s$  can be divided into a fixed part  $R_2$  and a variable part  $(R_2/s - R_2)$  i.e.,

$$\frac{R_2}{s} = R_2 + R_2\left(\frac{1}{s} - 1\right)$$

- (i) The first part  $R_2$  is the rotor resistance/phase, and represents the rotor Cu loss.
- (ii) The second part  $R_2\left(\frac{1}{s}-1\right)$  is a variable-resistance load. The power delivered to this load represents the total mechanical power developed in the rotor. Thus mechanical load on the induction motor can be replaced by a variable-resistance load of value  $R_2\left(\frac{1}{s}-1\right)$ . This is

$$\therefore R_L = R_2\left(\frac{1}{s}-1\right)$$

Fig. 3.10 (iii) shows the equivalent rotor circuit along with load resistance  $R_L$ .

Now Fig: 3.11 shows the equivalent circuit per phase of a 3-phase induction motor. Note that mechanical load on the motor has been replaced by an equivalent electrical resistance  $R_L$  given by;

$$R_L = R_2\left(\frac{1}{s}-1\right) \quad \text{----- (i)}$$

The circuit shown in Fig. 3.11 is similar to the equivalent circuit of a transformer with secondary load equal to  $R_2$  given by eq. (i). The rotor e.m.f. in the equivalent circuit now depends only on the transformation ratio  $K (= E_2/E_1)$ .

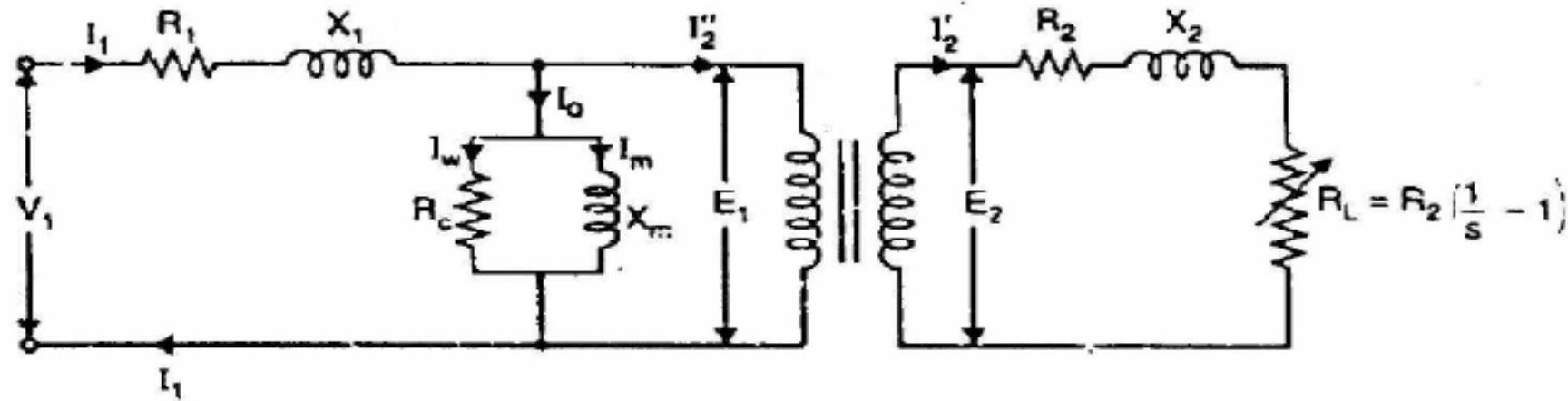


Fig: 3.11

Therefore; induction motor can be represented as an equivalent transformer connected to a variable-resistance load  $R_L$  given by eq. (i). The power delivered to  $R_L$  represents the total mechanical power developed in the rotor. Since the equivalent circuit of Fig. 3.11 is that of a transformer, the secondary (i.e., rotor) values can be transferred to primary (i.e., stator) through the appropriate use of transformation ratio  $K$ . Recall that when shifting resistance/reactance from secondary to primary, it should be divided by  $K^2$  whereas current should be multiplied by  $K$ . The equivalent circuit of an induction motor referred to primary is shown in Fig. 3.12.

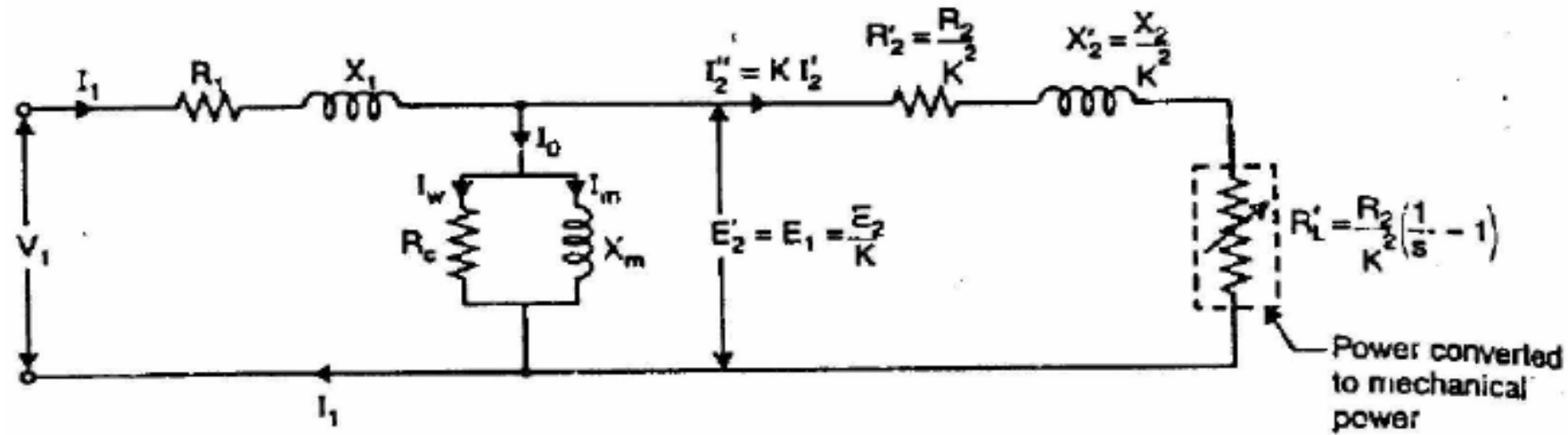


Fig: 3.12

Note that the element (i.e.,  $R_L'$ ) enclosed in the dotted box is the equivalent electrical resistance related to the mechanical load on the motor. The following points may be noted from the equivalent circuit of the induction motor:

(i) At no-load, the slip is practically zero and the load  $R_L'$  is infinite. This condition resembles that in a transformer whose secondary winding is open-circuited.

(ii) At standstill, the slip is unity and the load  $R'_L$  is zero. This condition resembles that in a transformer whose secondary winding is short-circuited.

(iii) When the motor is running under load, the value of  $R'_L$  will depend upon the value of the slip  $s$ . This condition resembles that in a transformer whose secondary is supplying variable and purely resistive load.

(iv) The equivalent electrical resistance  $R'_L$  related to mechanical load is slip or speed dependent. If the slip  $s$  increases, the load  $R'_L$  decreases and the rotor current increases and motor will develop more mechanical power. This is expected because the slip of the motor increases with the increase of load on the motor shaft.

### **3.8 Power and Torque Relations of Three Phase Induction Motor**

The transformer equivalent circuit of an induction motor is quite helpful in analyzing the various power relations in the motor. Fig. 3.13 shows the equivalent circuit per phase of an induction motor where all values have been referred to primary (i.e., stator).

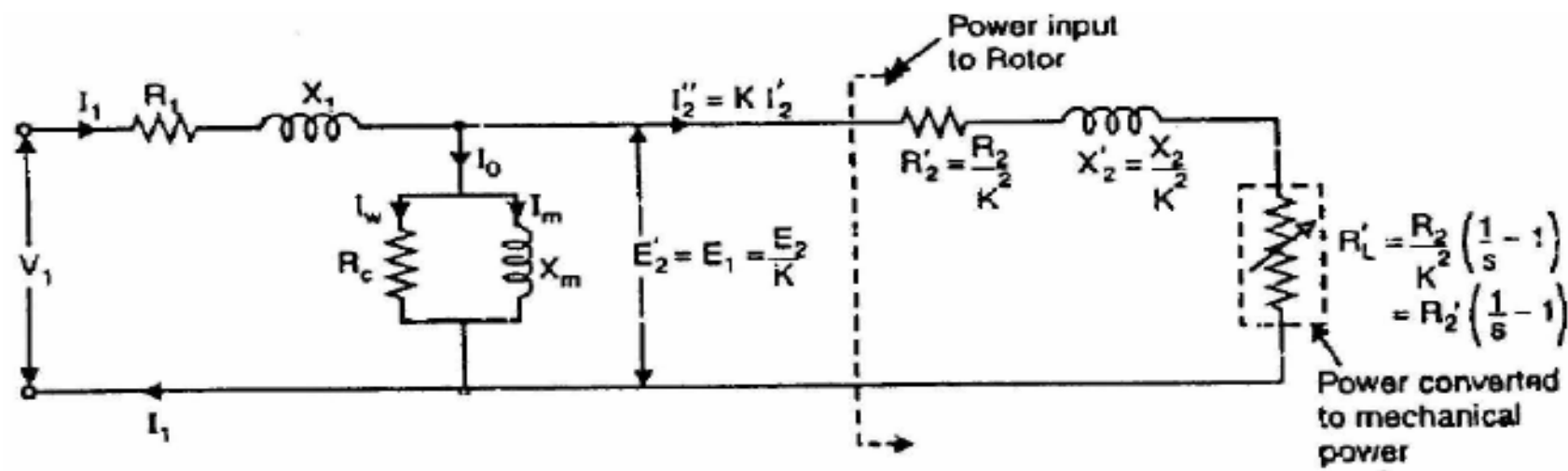


Fig: 3.13

(i) Total electrical load  $= R'_2 \left( \frac{1}{s} - 1 \right) + R'_2 = \frac{R'_2}{s}$

Power input to stator  $= 3V_1 I_1 \cos \phi_1$

There will be stator core loss and stator Cu loss. The remaining power will be the power transferred across the air-gap i.e., input to the rotor.

(ii) Rotor input  $= \frac{3(I'_2)^2 R'_2}{s}$

Rotor Cu loss  $= 3(I'_2)^2 R'_2$

Total mechanical power developed by the rotor is

$P_m = \text{Rotor input} - \text{Rotor Cu loss}$

$$= \frac{3(I'_2)^2 R'_2}{s} - 3(I'_2)^2 R'_2 = 3(I'_2)^2 R'_2 \left( \frac{1}{s} - 1 \right)$$

(iii) If  $T_g$  is the gross torque developed by the rotor, then,

$$P_m = \frac{2\pi N T_g}{60}$$

$$\text{or} \quad 3(I'_2)^2 R'_2 \left( \frac{1}{s} - 1 \right) = \frac{2\pi N T_g}{60}$$

$$\text{or} \quad 3(I'_2)^2 R'_2 \left( \frac{1-s}{s} \right) = \frac{2\pi N T_g}{60}$$

$$\text{or} \quad 3(I'_2)^2 R'_2 \left( \frac{1-s}{s} \right) = \frac{2\pi N_s (1-s) T_g}{60} \quad [\because N = N_s (1-s)]$$

$$\therefore T_g = \frac{3(I'_2)^2 R'_2 / s}{2\pi N_s / 60} \quad \text{N - m}$$

$$\text{or} \quad T_g = 9.55 \frac{3(I'_2)^2 R'_2 / s}{N_s} \quad \text{N - m}$$

Note that shaft torque  $T_{sh}$  will be less than  $T_g$  by the torque required to meet windage and frictional losses.

### 3.10 Rotor Output

If  $T_g$  newton-metre is the gross torque developed and  $N$  r.p.m. is the speed of the rotor, then,

$$\text{Gross rotor output} = \frac{2\pi N T_g}{60} \text{ watts}$$

If there were no copper losses in the rotor, the output would equal rotor input and the rotor would run at synchronous speed  $N_s$ .

$$\therefore \text{Rotor input} = \frac{2\pi N_s T_g}{60} \text{ watts}$$

$$\begin{aligned} \therefore \text{Rotor Cu loss} &= \text{Rotor input} - \text{Rotor output} \\ &= \frac{2\pi T_g}{60} (N_s - N) \end{aligned}$$

$$(i) \quad \frac{\text{Rotor Cu loss}}{\text{Rotor input}} = \frac{N_s - N}{N_s} = s$$

$$\therefore \text{Rotor Cu loss} = s \times \text{Rotor input}$$

$$\begin{aligned} \text{(ii) Gross rotor output, } P_m &= \text{Rotor input} - \text{Rotor Cu loss} \\ &= \text{Rotor input} - s \times \text{Rotor input} \end{aligned}$$

$$\therefore P_m = \text{Rotor input} (1 - s)$$

$$\text{(iii) } \frac{\text{Gross rotor output}}{\text{Rotor input}} = 1 - s = \frac{N}{N_s}$$

$$\text{(iv) } \frac{\text{Rotor Cu loss}}{\text{Gross rotor output}} = \frac{s}{1 - s}$$

It is clear that if the input power to rotor is “Pr” then “s.Pr” is lost as rotor Cu loss and the remaining (1 - s) Pr is converted into mechanical power. Consequently, induction motor operating at high slip has poor efficiency.

*Note.*

$$\frac{\text{Gross rotor output}}{\text{Rotor input}} = 1 - s$$

If the stator losses as well as friction and windage losses are neglected, then,

$$\text{Gross rotor output} = \text{Useful output}$$

$$\text{Rotor input} = \text{Stator input}$$

$$\therefore \frac{\text{Useful output}}{\text{Stator input}} = 1 - s = \text{Efficiency}$$

Hence the approximate efficiency of an induction motor is  $1 - s$ . Thus if the slip of an induction motor is 0.125, then its approximate efficiency is  $= 1 - 0.125 = 0.875$  or 87.5%.

### 3.11.1 Torque Equations

The gross torque  $T_g$  developed by an induction motor is given by;

$$\begin{aligned} T_g &= \frac{\text{Rotor input}}{2\pi N_s} && \dots N_s \text{ is r.p.s.} \\ &= \frac{60 \times \text{Rotor input}}{2\pi N_s} && \dots N_s \text{ is r.p.s.} \end{aligned}$$

$$\text{Now Rotor input} = \frac{\text{Rotor Cu loss}}{s} = \frac{3(I'_2)^2 R_2}{s} \quad (i)$$

As shown in Sec. 8.16, under running conditions,

$$I'_2 = \frac{s E_2}{\sqrt{R_2^2 + (s X_2)^2}} = \frac{s K E_1}{\sqrt{R_2^2 + (s X_2)^2}}$$

$$\text{where } K = \text{Transformation ratio} = \frac{\text{Rotor turns/phase}}{\text{Stator turns/phase}}$$

$$\therefore \text{Rotor input} = 3 \times \frac{s^2 E_2^2 R_2}{R_2^2 + (s X_2)^2} \times \frac{1}{s} = \frac{3 s E_2^2 R_2}{R_2^2 + (s X_2)^2}$$

(Putting the value of  $I_2$  in eq.(i))

Also 
$$\text{Rotor input} = 3 \times \frac{s^2 K^2 E_1^2 R_2}{R_2^2 + (s X_2)^2} \times \frac{1}{s} = \frac{3 s K^2 E_1^2 R_2}{R_2^2 + (s X_2)^2}$$

(Putting the value of  $I_2$  in eq.(i))

$$\therefore T_g = \frac{\text{Rotor input}}{2\pi N_s} = \frac{3}{2\pi N_s} \times \frac{s E_2^2 R_2}{R_2^2 + (s X_2)^2} \quad \dots \text{in terms of } E_2$$

$$= \frac{3}{2\pi N_s} \times \frac{s K^2 E_1^2 R_2}{R_2^2 + (s X_2)^2} \quad \dots \text{in terms of } E_1$$

Note that in the above expressions of  $T_g$ , the values  $E_1$ ,  $E_2$ ,  $R_2$  and  $X_2$  represent the phase values.

### 3.11.2 Rotor Torque

The torque  $T$  developed by the rotor is directly proportional to:

- (i) rotor current
- (ii) rotor e.m.f.
- (iii) power factor of the rotor circuit

$$\therefore T \propto E_2 I_2 \cos \phi_2$$

or  $T = K E_2 I_2 \cos \phi_2$

where  $I_2$  = rotor current at standstill  
 $E_2$  = rotor e.m.f. at standstill  
 $\cos \phi_2$  = rotor p.f. at standstill

**Note.** The values of rotor e.m.f., rotor current and rotor power factor are taken for the given conditions.

### 3.11.3 Starting Torque ( $T_s$ )

Let,

$E_2$  = rotor e.m.f. per phase at standstill

$X_2$  = rotor reactance per phase at standstill

$R_2$  = rotor resistance per phase

Rotor impedance/phase,  $Z_2 = \sqrt{R_2^2 + X_2^2}$  ...at standstill

Rotor current/phase,  $I_2 = \frac{E_2}{Z_2} = \frac{E_2}{\sqrt{R_2^2 + X_2^2}}$  ...at standstill

Rotor p.f.,  $\cos \phi_2 = \frac{R_2}{Z_2} = \frac{R_2}{\sqrt{R_2^2 + X_2^2}}$  ...at standstill

$$\begin{aligned}\therefore \text{ Starting torque, } T_s &= K E_2 I_2 \cos \phi_2 \\ &= K E_2 \times \frac{E_2}{\sqrt{R_2^2 + X_2^2}} \times \frac{R_2}{\sqrt{R_2^2 + X_2^2}} \\ &= \frac{K E_2^2 R_2}{R_2^2 + X_2^2}\end{aligned}$$

Generally, the stator supply voltage  $V$  is constant so that flux per pole  $\phi$  set up by the stator is also fixed. This in turn means that e.m.f.  $E_2$  induced in the rotor will be constant.

$$\therefore T_s = \frac{K_1 R_2}{R_2^2 + X_2^2} = \frac{K_1 R_2}{Z_2^2}$$

where  $K_1$  is another constant.

It is clear that the magnitude of starting torque would depend upon the relative values of  $R_2$  and  $X_2$  i.e., rotor resistance/phase and standstill rotor reactance/phase.

It can be shown that  $K = 3/2 \pi N_s$ .

$$\therefore T_s = \frac{3}{2\pi N_s} \cdot \frac{E_2^2 R_2}{R_2^2 + X_2^2}$$

Note that here  $N_s$  is in r.p.s.

### 3.11.4 Condition for Maximum Starting Torque

It can be proved that starting torque will be maximum when rotor resistance/phase is equal to standstill rotor reactance/phase.

$$\text{Now } T_s = \frac{K_1 R_2}{R_2^2 + X_2^2} \quad (i)$$

Differentiating eq. (i) w.r.t.  $R_2$  and equating the result to zero, we get,

$$\frac{dT_s}{dR_2} = K_1 \left[ \frac{1}{R_2^2 + X_2^2} - \frac{R_2(2R_2)}{(R_2^2 + X_2^2)^2} \right] = 0$$

$$\text{or } R_2^2 + X_2^2 = 2R_2^2$$

$$\text{or } R_2 = X_2$$

Hence starting torque will be maximum when:

$$\text{Rotor resistance/phase} = \text{Standstill rotor reactance/phase}$$

Under the condition of maximum starting torque,  $\phi_2 = 45^\circ$  and rotor power factor is 0.707 lagging.

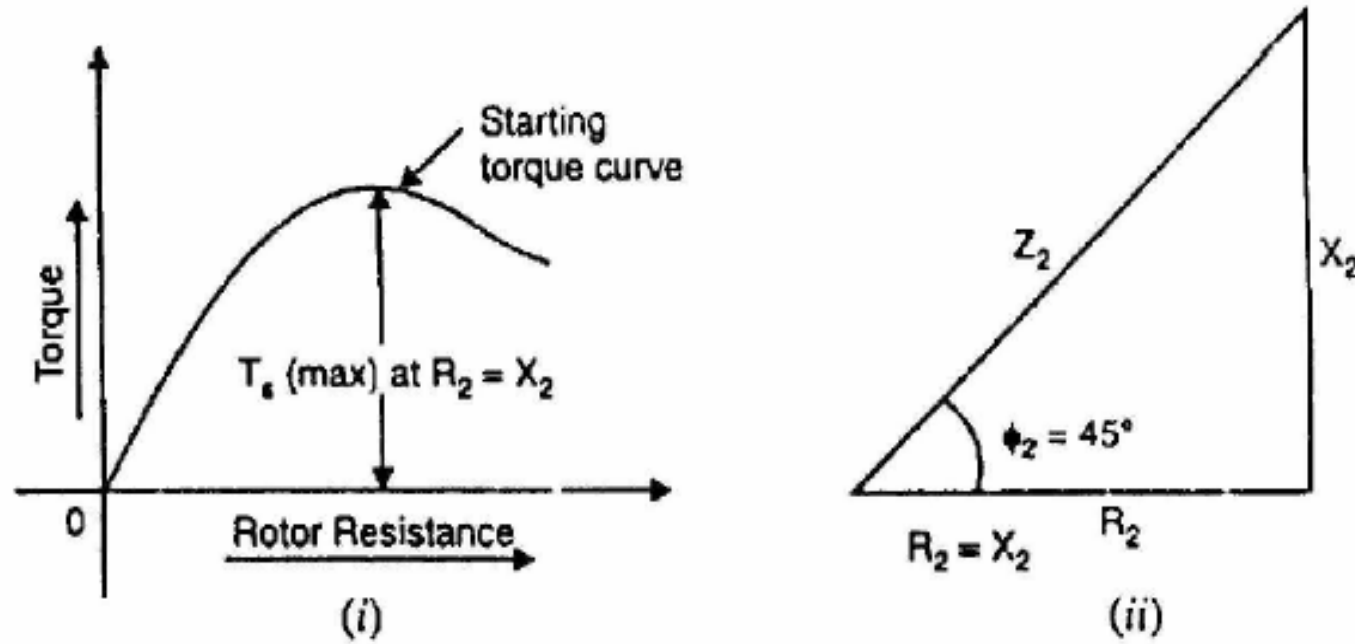


Fig: 3.14

Fig. 3.14 shows the variation of starting torque with rotor resistance. As the rotor resistance is increased from a relatively low value, the starting torque increases until it becomes maximum when  $R_2 = X_2$ . If the rotor resistance is increased beyond this optimum value, the starting torque will decrease.

### 3.11.5 Effect of Change of Supply Voltage

$$T_s = \frac{K E_2^2 R_2}{R_2^2 + X_2^2}$$

Since  $E_2 \propto$  Supply voltage  $V$

$$\therefore T_s = \frac{K_2 V^2 R_2}{R_2^2 + X_2^2}$$

where  $K_2$  is another constant.

$$\therefore T_s \propto V^2$$

Therefore, the starting torque is very sensitive to changes in the value of supply voltage. For example, a drop of 10% in supply voltage will decrease the starting torque by about 20%. This could mean the motor failing to start if it cannot produce a torque greater than the load torque plus friction torque.

### 3.12 Circle Diagram

To analyse the three phase induction motor performance using circle diagram we need to determine the equivalent circuit parameters of the machine.

#### 3.12.1 Approximate Equivalent Circuit of Induction Motor

As in case of a transformer, the approximate equivalent circuit of an induction motor is obtained by shifting the shunt branch ( $R_c - X_m$ ) to the input terminals as shown in Fig. 3.15. This step has been taken on the assumption that voltage drop in  $R_1$  and  $X_1$  is small and the terminal voltage  $V_1$  does not appreciably differ from the induced voltage  $E_1$ . Fig. 3.15 shows the approximate equivalent circuit per phase of an induction motor where all values have been referred to primary (i.e., stator).

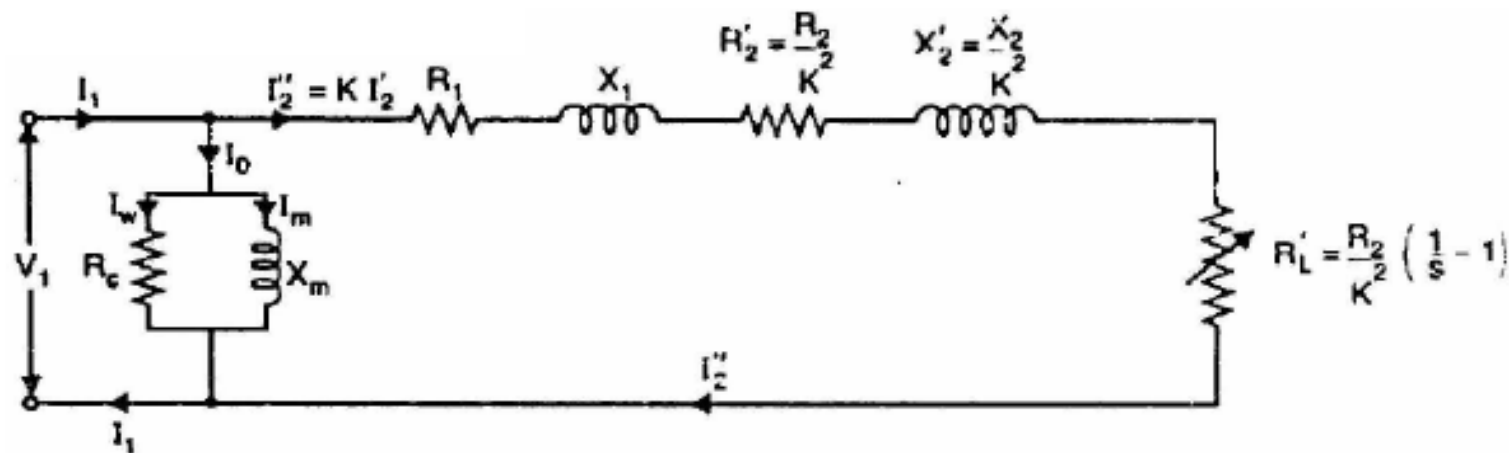


Fig: 3.15

The above approximate circuit of induction motor is not so readily justified as with the transformer. This is due to the following reasons:

- (i) Unlike that of a power transformer, the magnetic circuit of the induction motor has an air-gap. Therefore, the exciting current of induction motor (30 to 40% of full-load current) is much higher than that of the power transformer. Consequently, the exact equivalent circuit must be used for accurate results.
- (ii) The relative values of  $X_1$  and  $X_2$  in an induction motor are larger than the corresponding ones to be found in the transformer. This fact does not justify the use of approximate equivalent circuit
- (iii) In a transformer, the windings are concentrated whereas in an induction motor, the windings are distributed. This affects the transformation ratio.

In spite of the above drawbacks of approximate equivalent circuit, it yields results that are satisfactory for large motors. However, approximate equivalent circuit is not justified for small motors.

### 3.12.2 Tests to Determine the Equivalent Circuit Parameters

In order to find values for the various elements of the equivalent circuit, tests must be conducted on a particular machine, which is to be represented by the equivalent circuit. In order to do this, we note the following.

1. When the machine is run on no-load, there is very little torque developed by it. In an ideal case where there is no mechanical losses, there is no mechanical power developed at no-load. Recalling the explanations in the section on torque production, the flow of current in the rotor is indicative of the torque that is produced. If no torque is produced, one may conclude that no current would be flowing in the rotor either. The rotor branch acts like an open circuit. This conclusion may also be reached by reasoning that when there is no load, an ideal machine will run up to its synchronous speed where the slip is zero resulting in an infinite impedance in the rotor branch.

2. When the machine is prevented from rotation, and supply is given, the slip remains at unity. The elements representing the magnetizing branch  $R_m$  &  $X_m$  are high impedances much larger than  $R'_r$  &  $X'_{lr}$  in series. Thus, in the exact equivalent circuit of the induction machine, the magnetizing branch may be neglected.

From these considerations, we may reduce the induction machine equivalent circuit of Fig.3.13 & Fig: 3.15 to those shown in Fig: 3.16.

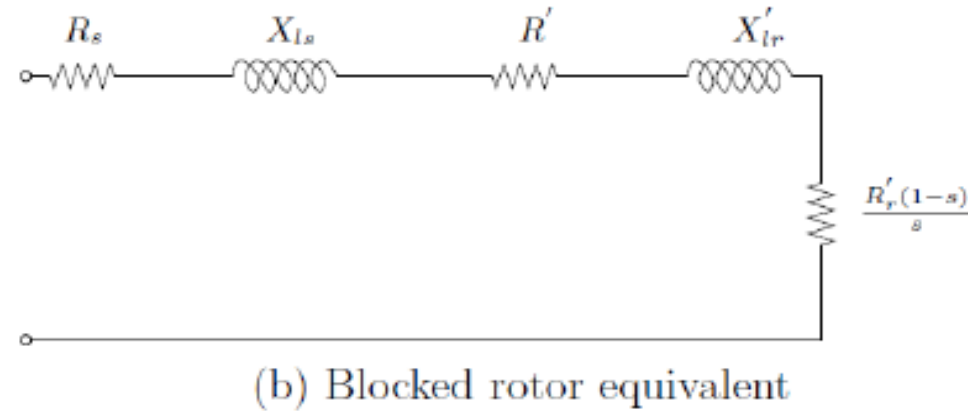
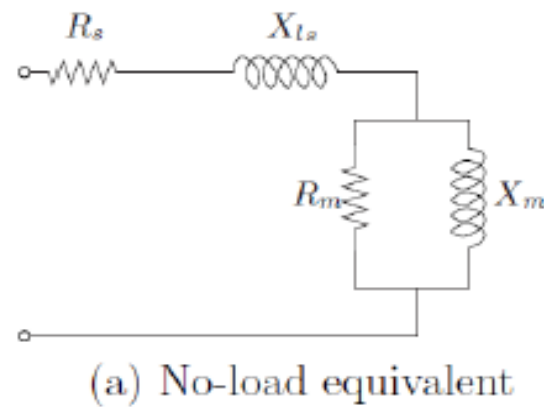


Fig: 3.16

These two observations and the reduced equivalent circuits are used as the basis for the two most commonly used tests to find out the equivalent circuit parameters — the blocked rotor test and no load test. They are also referred to as the short circuit test and open circuit test respectively in conceptual analogy to the transformer.

### 1. No-load test

The behaviour of the machine may be judged from the equivalent circuit of Fig: 3.16 (a). The current drawn by the machine causes a stator-impedance drop and the balance voltage is applied across the magnetizing branch. However, since the magnetizing branch impedance is large, the current drawn is small and hence the stator impedance drop is small compared to the applied voltage (rated value). This drop and the power dissipated in the stator resistance are therefore neglected and the total power drawn is assumed to be consumed entirely as core loss. This can also be seen from the approximate equivalent circuit, the use of which is justified by the foregoing arguments. This test therefore enables us to compute the resistance and inductance of the magnetizing branch in the following manner.

Let applied voltage =  $V_s$ . Then current drawn is given by

$$I_s = \frac{V_s}{R_m} + \frac{V_s}{jX_m}$$

The power drawn is given by

$$P_s = \frac{V_s^2}{R_m} \Rightarrow R_m = \frac{V_s^2}{P_s}$$

$V_s$ ,  $I_s$  and  $P_s$  are measured with appropriate meters. With  $R_m$  known by above equation,  $X_m$  also can be found. The current drawn is at low power factor and hence a suitable wattmeter should be used.

## 2. Blocked-rotor Test

In this test the rotor is prevented from rotation by mechanical means and hence the name. Since there is no rotation, slip of operation is unity,  $s = 1$ . The equivalent circuit valid under these conditions is shown in Fig: 3.16 (b). Since the current drawn is decided by the resistance and leakage impedances alone, the magnitude can be very high when rated voltage is applied. Therefore in this test, only small voltages are applied — just enough to cause rated current to flow. While the current magnitude depends on the resistance and the reactance, the power drawn depends on the resistances.

The parameters may then be determined as follows. The source current and power drawn may be written as -

$$I_s = \frac{V_s}{(R_s + R'_r) + j(X_s + X'_r)}$$
$$P_s = |I_s|^2 (R_s + R'_r)$$

In the test  $V_s$ ,  $I_s$  and  $P_s$  are measured with appropriate meters. Above equation enables us to compute  $(R_s + R'_r)$ . Once this is known,  $(X_s + X'_r)$  may be computed from the above equation.

Note that this test only enables us to determine the series combination of the resistance and the reactance only and not the individual values. Generally, the individual values are assumed to be equal; the assumption  $R_s = R'_r$ , and  $X_s = X'_r$  suffices for most purposes.

In practice, there are differences. If more accurate estimates are required IEEE guidelines may be followed which depend on the size of the machine.

These two tests determine the equivalent circuit parameters in a 'Stator-referred' sense, i.e., the rotor resistance and leakage inductance are not the actual values but what they 'appear to be' when looked at from the stator. This is sufficient for most purposes as interconnections to the external world are generally done at the stator terminals.

### 3.12.3 Construction of Circle Diagram

Conduct No load test and blocked rotor test on the induction motor and find out the per phase values of no load current  $I_0$ , short circuit current  $I_{sc}$  and the corresponding phase angles  $\Phi_0$  and  $\Phi_{sc}$ . Also find short circuit current  $I_{SN}$  corresponding to normal supply voltage. With this data, the circle diagram can be drawn as follows see Fig: 3.17.

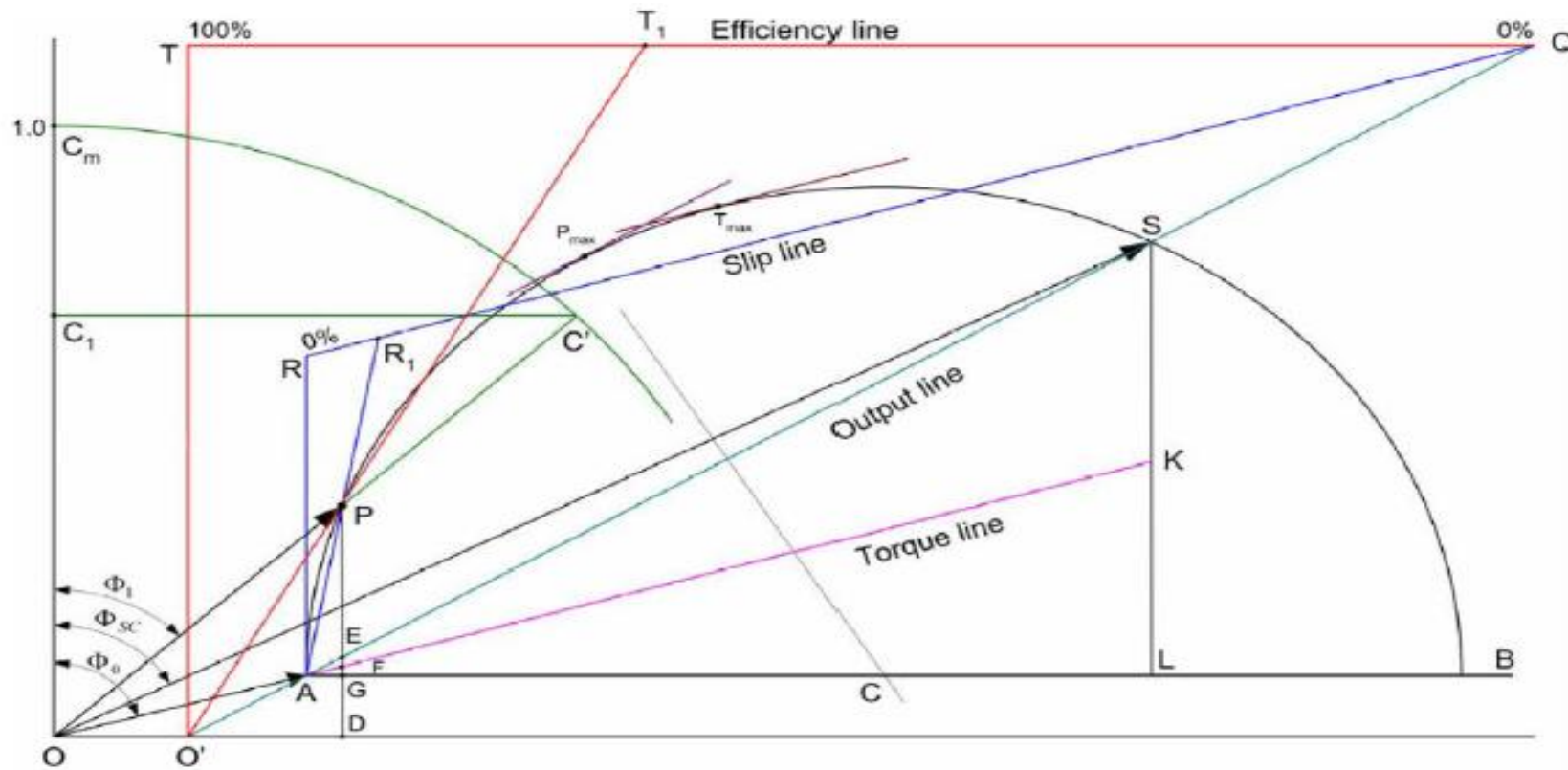


Fig: 3.17 Construction of Circle Diagram

1. With suitable scale, draw vector OA with length corresponding to  $I_0$  at an angle  $\Phi_0$  from the vertical axis. Draw a horizontal line AB.
2. Draw OS equal to  $I_{SN}$  at an angle  $\Phi_{SC}$  and join AS.
3. Draw the perpendicular bisector to AS to meet the horizontal line AB at C.
4. With C as centre, draw a portion of circle passing through A and S. This forms the circle diagram which is the locus of the input current.
5. From point S, draw a vertical line SL to meet the line AB.
6. Divide SL at point K so that  $SK : KL = \text{rotor resistance} : \text{stator resistance}$ .
7. For a given operating point P, draw a vertical line PEF GD as shown. then PE = output power, EF = rotor copper loss, FG = stator copper loss, GD = constant loss (iron loss + mechanical loss)
8. To find the operating points corresponding to maximum power and maximum torque, draw tangents to the circle diagram parallel to the output line and torque line respectively. The points at which these tangents touch the circle are respectively the maximum power point and maximum torque point.

### *Efficiency line*

1. The output line AS is extended backwards to meet the X-axis at O'.
2. From any convenient point on the extended output line, draw a horizontal line QT so as to meet the vertical from O'. Divide the line QT into 100 equal parts.
3. To find the efficiency corresponding to any operating point P, draw a line from O' to the efficiency line through P to meet the efficiency line at T1. Now QT1 is the efficiency.

### *Slip Line*

1. Draw line QR parallel to the torque line, meeting the vertical through A at R. Divide RQ into 100 equal parts.
2. To find the slip corresponding to any operating point P, draw a line from A to the slip line through P to meet the slip line at R1. Now RR1 is the slip

### *Power Factor Curve*

1. Draw a quadrant of a circle with O as centre and any convenient radius. Divide OCm into 100 equal parts.
2. To find power factor corresponding to P, extend the line OP to meet the power factor curve at C'. Draw a horizontal line C'C1 to meet the vertical axis at C1. Now OC1 represents power factor.

### 3.13 Performance Characteristics of Three phase Induction Motor

The equivalent circuits derived in the preceding section can be used to predict the performance characteristics of the induction machine. The important performance characteristics in the steady state are the efficiency, power factor, current, starting torque, maximum (or pull-out) torque.

#### 3.13.1 The complete torque-speed characteristic

In order to estimate the speed torque characteristic let us suppose that a sinusoidal voltage is impressed on the machine. Recalling that the equivalent circuit is the per-phase representation of the machine, the current drawn by the circuit is given by

$$I_s = \frac{V_s}{(R_s + \frac{R'_r}{s}) + j(X_{ls} + X'_{lr})}$$

Where,  $V_s$  is the phase voltage phasor and  $I_s$  is the current phasor. The magnetizing current is neglected. Since this current is flowing through  $R'_r/s$ , the air-gap power is given by

$$\begin{aligned} P_g &= |I_s|^2 \frac{R'_r}{s} \\ &= \frac{V_s^2}{(R_s + \frac{R'_r}{s})^2 + (X_{ls} + X'_{lr})^2} \frac{R'_r}{s} \end{aligned}$$

The mechanical power output was shown to be  $(1-s)P_g$  (power dissipated in  $R'_r/s$ ). The torque is obtained by dividing this by the shaft speed  $\omega_m$ . Thus we have,

$$\frac{P_g(1-s)}{\omega_m} = \frac{P_g(1-s)}{\omega_s(1-s)} = |I_s|^2 \frac{R'_r}{s\omega_s}$$

where  $\omega_m$  is the synchronous speed in radians per second and  $s$  is the slip. Further, this is the torque produced per phase. Hence the overall torque is given by

$$T_e = \frac{3}{\omega_s} \cdot \frac{V_s^2}{(R_s + \frac{R'_r}{s})^2 + (X_{ls} + X'_{lr})^2} \cdot \frac{R'_r}{s}$$

The torque may be plotted as a function of 's' and is called the torque-slip (or torque-speed, since slip indicates speed) characteristic a very important characteristic of the induction machine.

A typical torque-speed characteristic is shown in Fig: 3.18. This plot corresponds to a 3 kW, 4 pole, and 60 Hz machine. The rated operating speed is 1780 rpm.

Further, this curve is obtained by varying slip with the applied voltage being held constant. Coupled with the fact that this is an equivalent circuit valid under steady state, it implies that if this characteristic is to be measured experimentally, we need to look at the torque for a given speed after all transients have died down. One cannot, for example, try to obtain this curve by directly starting the motor with full voltage applied to the terminals and measuring the torque and speed dynamically as it runs up to steady speed.

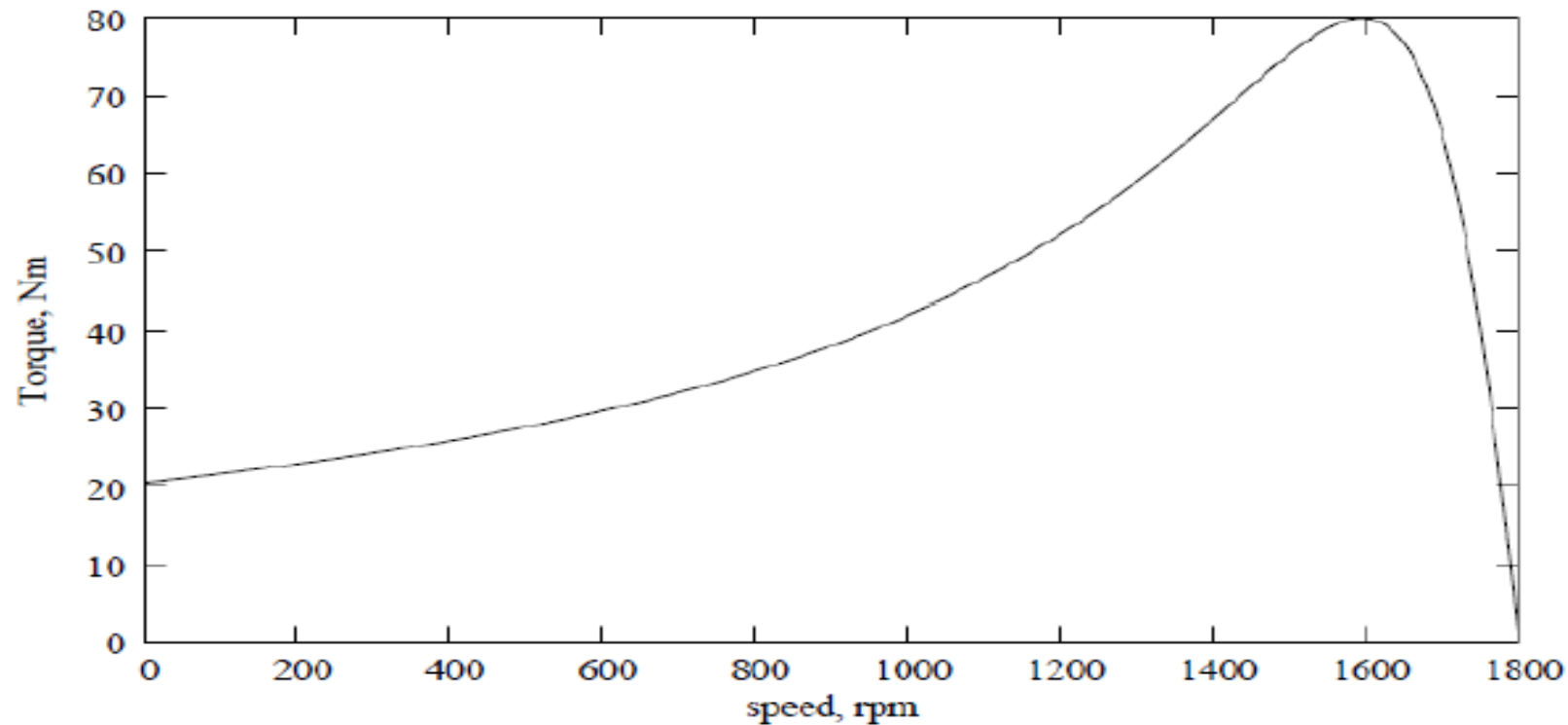


Fig: 3.18

With respect to the direction of rotation of the air-gap flux, the rotor may be driven to higher speeds by a prime mover or may also be rotated in the reverse direction. The torque-speed relation for the machine under the entire speed range is called the complete speed-torque characteristic. A typical curve is shown in Fig: 3.19 for a four-pole machine, the synchronous speed being 1500 rpm. Note that negative speeds correspond to slip values greater than 1, and speeds greater than 1500 rpm correspond to negative slip. The plot also shows the operating modes of the induction machine in various regions. The slip axis is also shown for convenience.

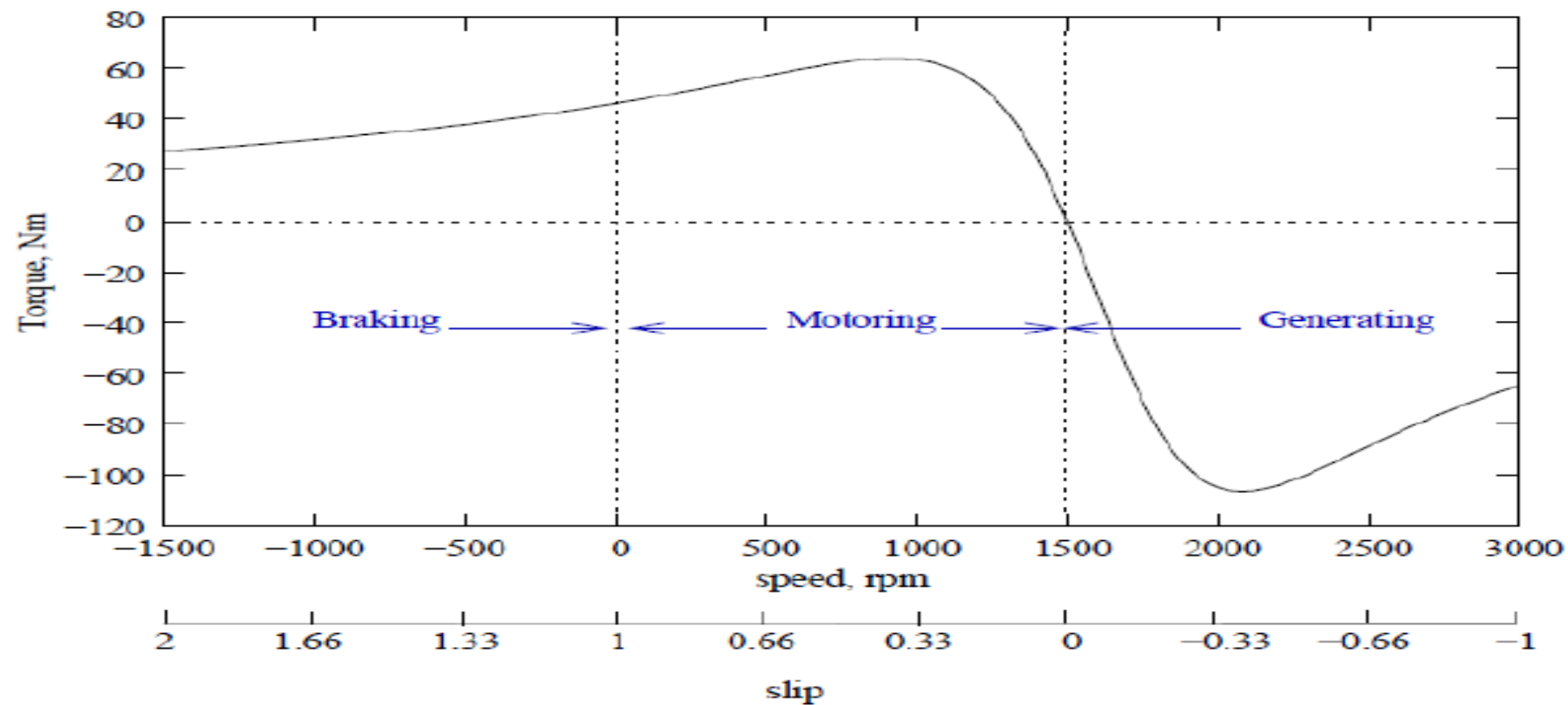


Fig: 3.19

### 3.13.2 Effect of Rotor Resistance on Speed Torque Characteristic

Restricting ourselves to positive values of slip, we see that the curve has a peak point. This is the maximum torque that the machine can produce, and is called as stalling torque. If the load torque is more than this value, the machine stops rotating or stalls. It occurs at a slip  $\hat{s}$ , which for the machine of Fig: 3.19 is 0.38. At values of slip lower than  $\hat{s}$ , the curve falls steeply down to zero at  $s = 0$ . The torque at synchronous speed is therefore zero. At values of slip higher than  $s = \hat{s}$ , the curve falls slowly to a minimum value at  $s = 1$ . The torque at  $s = 1$  (speed = 0) is called the starting torque. The value of the stalling torque may be obtained by differentiating the expression for torque with respect to zero and setting it to zero to find the value of  $\hat{s}$ . Using this method, we can write –

$$\hat{s} = \frac{\pm R'_r}{\sqrt{R_s'^2 + (X_{ls} + X'_{lr})^2}}$$

Substituting  $\hat{s}$  into the expression for torque gives us the value of the stalling torque  $\hat{T}_e$ ,

$$\hat{T}_e = \frac{3V_s^2}{2\omega_s} \cdot \frac{1}{R_s \pm \sqrt{R_s'^2 + (X_{ls} + X'_{lr})^2}}$$

- The negative sign being valid for negative slip.

The expression shows that  $\hat{T}_e$  is independent of  $R'r$ , while  $\hat{s}$  is directly proportional to  $R'r$ . This fact can be made use of conveniently to alter  $\hat{s}$ . If it is possible to change  $R'r$ , then we can get a whole series of torque-speed characteristics, the maximum torque remaining constant all the while.

We may note that if  $R'r$  is chosen equal to =

$$\sqrt{R_s^2 + (X_{ls} + X_{lr}')^2}$$

The  $\hat{s}$ , becomes unity, which means that the maximum torque occurs at starting. Thus changing of  $R'r$ , wherever possible can serve as a means to control the starting torque Fig: 3.20.

While considering the negative slip range, (generator mode) we note that the maximum torque is higher than in the positive slip region (motoring mode).

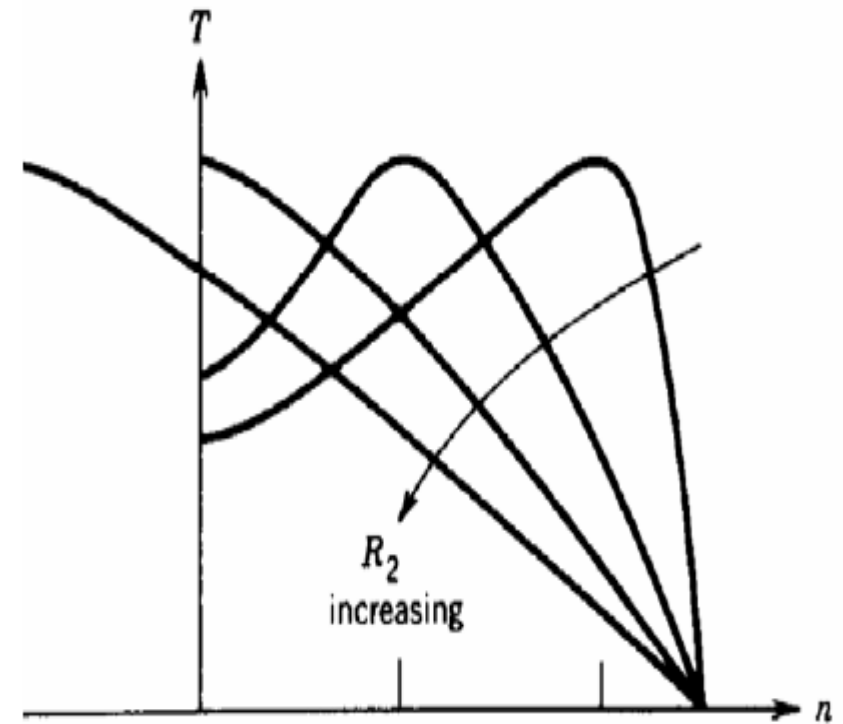


Fig: 3.20

### 3.13.3 Operating Point and Stable & Unstable region of Operation

Consider a speed torque characteristic shown in fig. 25 for an induction machine, having the load characteristic also superimposed on it. The load is a constant torque load i.e. the torque required for operation is fixed irrespective of speed.

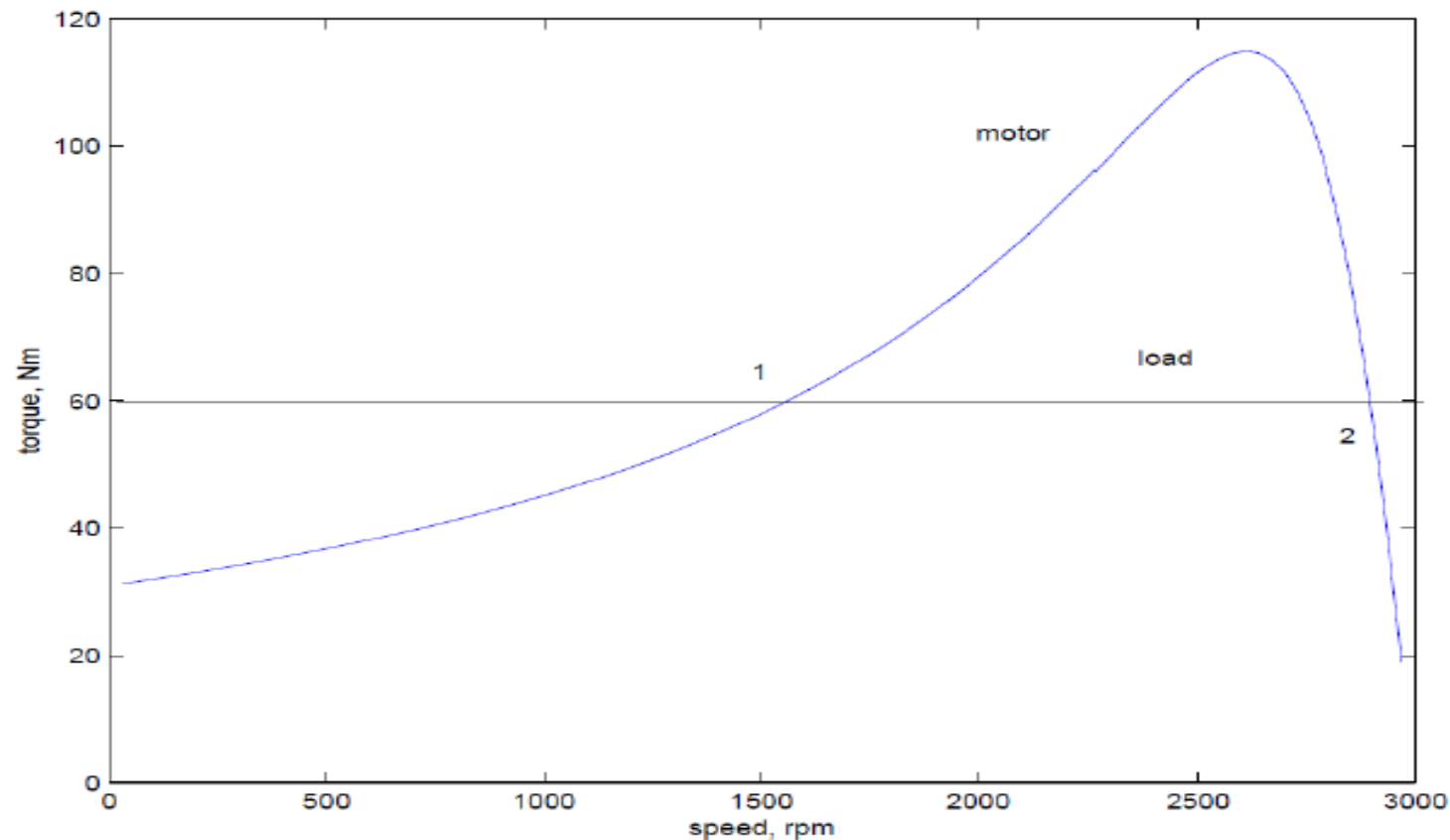


Fig: 3.21

The system consisting of the motor and load will operate at a point where the two characteristics meet. From the above plot, we note that there are two such points. We therefore need to find out which of these is the actual operating point. To answer this we must note that, in practice, the characteristics are never fixed; they change slightly with time. It would be appropriate to consider a small band around the curve drawn where the actual points of the characteristic will lie. This being the case let us consider that the system is operating at point 1, and the load torque demand increases slightly. This is shown in Fig: 3.22, where the change is exaggerated for clarity. This would shift the point of operation to a point 1' at which the slip would be less and the developed torque higher.

The difference in torque developed  $\Delta T_e$ , being positive will accelerate the machine. Any overshoot in speed as it approaches the point 1' will cause it to further accelerate since the developed torque is increasing. Similar arguments may be used to show that if for some reason the developed torque becomes smaller the speed would drop and the effect is cumulative. Therefore we may conclude that 1 is not a stable operating point.

Let us consider the point 2. If this point shifts to 2', the slip is now higher (speed is lower) and the positive difference in torque will accelerate the machine. This behaviour will tend to bring the operating point towards 2 once again. In other words, disturbances at point 2 will not cause a runaway effect. Similar arguments may be given for the case where the load characteristic shifts down. Therefore we conclude that point 2 is a stable operating point.

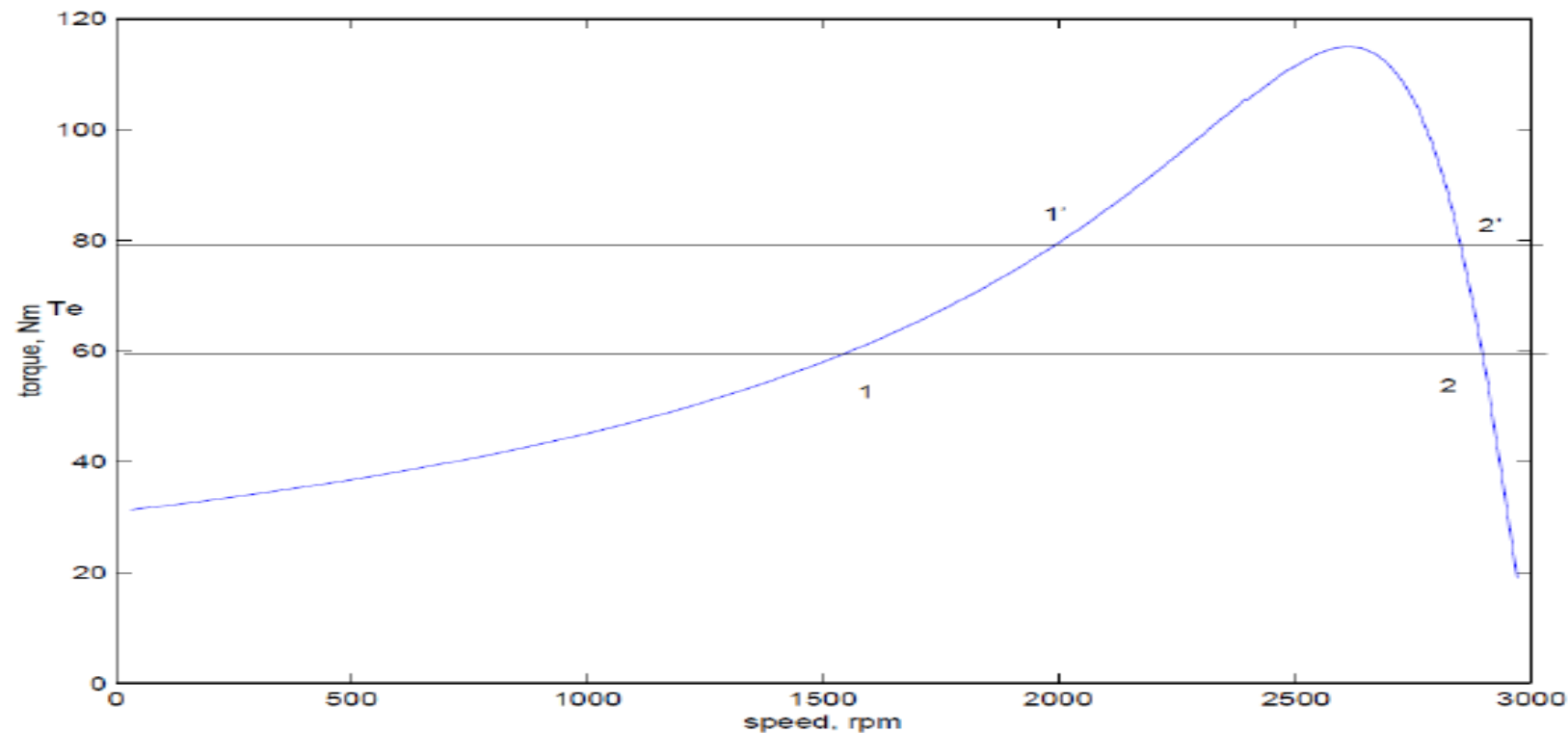


Fig: 3.22

From the above discussions, we can say that the entire region of the speed-torque characteristic from  $s = 0$  to  $s = \hat{s}$  is an unstable region, while the region from  $s = \hat{s}$  to  $s = 1$  is a stable region. Therefore the machine will always operate between  $s = 0$  and  $s = \hat{s}$ .

### 3.14 Operation with Unbalanced Supply Voltage on Polyphase Induction Motors

Three phase induction motors are designed and manufactured such that all three phases of the winding are carefully balanced with respect to the number of turns, placement of the winding, and winding resistance. When line voltages applied to a polyphase induction motor are not exactly the same, unbalanced currents will flow in the stator winding, the magnitude depending upon the amount of unbalance. A small amount of voltage unbalance may increase the current an excessive amount. The effect on the motor can be severe and the motor may overheat to the point of burnout.

#### *Unbalance Defined*

The voltage unbalance (or negative sequence voltage) in percent may be defined as follows:

$$\text{Percent Voltage Unbalance} = 100 * (\text{Maximum Voltage Deviation} / \text{Average Voltage})$$

### 3.15 Starting of Three Phase Induction Motor

The induction motor is fundamentally a transformer in which the stator is the primary and the rotor is short-circuited secondary. At starting, the voltage induced in the induction motor rotor is maximum ( $s = 1$ ). Since the rotor impedance is low, the rotor current is excessively large. This large rotor current is reflected in the stator because of transformer action. This results in high starting current (4 to 10 times the full-load current) in the stator at low power factor and consequently the value of starting torque is low. Because of the short duration, this value of large current does not harm the motor if the motor accelerates normally.

However, this large starting current will produce large line-voltage drop. This will adversely affect the operation of other electrical equipment connected to the same lines. Therefore, it is desirable and necessary to reduce the magnitude of stator current at starting and several methods are available for this purpose.

#### 3.15.1 Methods of Starting Three Phase Induction Motors

The method to be employed in starting a given induction motor depends upon the size of the motor and the type of the motor. The common methods used to start induction motors are:

- (i) Direct-on-line starting
- (ii) Stator resistance starting
- (iii) Autotransformer starting
- (iv) Star-delta starting
- (v) Rotor resistance starting

(i) *Direct-on-line starting*

This method of starting is just what the name implies—the motor is started by connecting it directly to 3-phase supply. The impedance of the motor at standstill is relatively low and when it is directly connected to the supply system, the starting current will be high (4 to 10 times the full load current) and at a low power factor. Consequently, this method of starting is suitable for relatively small (up to 7.5 kW) machines.

**Relation between starting and F.L. torques.** We know that:

$$\text{Rotor input} = 2\pi N_s T = kT$$

But Rotor Cu loss =  $s \times$  Rotor input

$$\therefore 3(I'_2)^2 R_2 = s \times kT$$

or  $T \propto (I'_2)^2 / s$

or  $T \propto I_1^2 / s$  ( $\because I'_2 \propto I_1$ )

If  $I_{st}$  is the starting current, then starting torque ( $T_{st}$ ) is

$$T \propto I_{st}^2 \quad (\because \text{at starting } s = 1)$$

If  $I_f$  is the full-load current and  $s_f$  is the full-load slip, then,

$$T_f \propto I_f^2 / s_f$$

$$\therefore \frac{T_{st}}{T_f} = \left( \frac{I_{st}}{I_f} \right)^2 \times s_f$$

When the motor is started direct-on-line, the starting current is the short-circuit (blocked-rotor) current  $I_{sc}$ .

$$\therefore \frac{T_{st}}{T_f} = \left( \frac{I_{sc}}{I_f} \right)^2 \times s_f$$

Let us illustrate the above relation with a numerical example. Suppose  $I_{sc} = 5 I_f$  and full-load slip  $s_f = 0.04$ . Then,

$$\frac{T_{st}}{T_f} = \left( \frac{I_{sc}}{I_f} \right)^2 \times s_f = \left( \frac{5 I_f}{I_f} \right)^2 \times 0.04 = (5)^2 \times 0.04 = 1$$

$$\therefore T_{st} = T_f$$

Note that starting current is as large as five times the full-load current but starting torque is just equal to the full-load torque. Therefore, starting current is very high and the starting torque is comparatively low. If this large starting current flows for a long time, it may overheat the motor and damage the insulation.

(ii) *Stator resistance starting*

In this method, external resistances are connected in series with each phase of stator winding during starting. This causes voltage drop across the resistances so that voltage available across motor terminals is reduced and hence the starting current. The starting resistances are gradually cut out in steps (two or more steps) from the stator circuit as the motor picks up speed. When the motor attains rated speed, the resistances are completely cut out and full line voltage is applied to the rotor see Fig: 3.23.

This method suffers from two drawbacks. First, the reduced voltage applied to the motor during the starting period lowers the starting torque and hence increases the accelerating time. Secondly, a lot of power is wasted in the starting resistances.

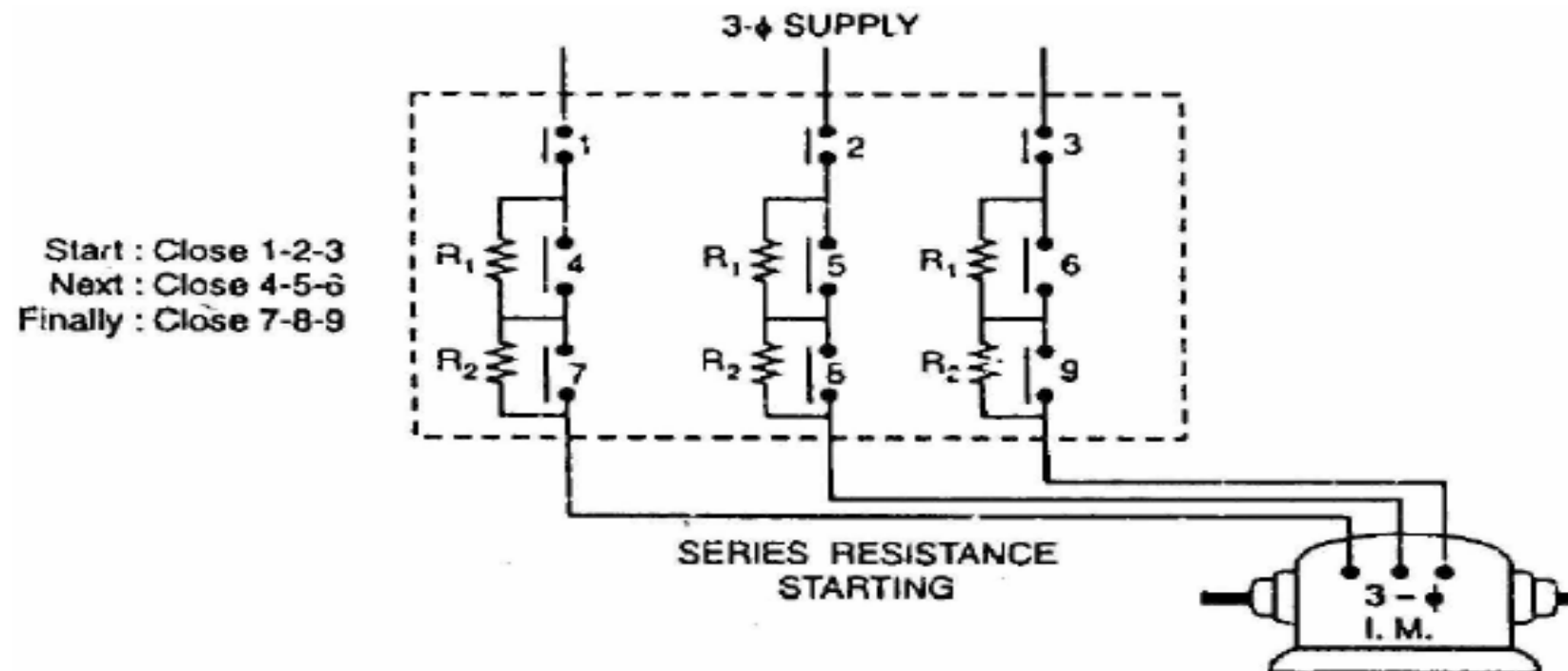


Fig: 3.23

### Relation between starting and F.L. torques.

Let  $V$  be the rated voltage/phase. If the voltage is reduced by a fraction  $x$  by the insertion of resistors in the line, then voltage applied to the motor per phase will be  $xV$ .

So,

$$I_{st} = x I_{sc}$$

Now 
$$\frac{T_{st}}{T_f} = \left( \frac{I_{st}}{I_f} \right)^2 \times s_f$$

or 
$$\frac{T_{st}}{T_f} = x^2 \left( \frac{I_{sc}}{I_f} \right)^2 \times s_f$$

Thus while the starting current reduces by a fraction  $x$  of the rated-voltage starting current ( $I_{sc}$ ), the starting torque is reduced by a fraction  $x^2$  of that obtained by direct switching. The reduced voltage applied to the motor during the starting period lowers the starting current but at the same time increases the accelerating time because of the reduced value of the starting torque. Therefore, this method is used for starting small motors only.

(iii) *Autotransformer starting*

This method also aims at connecting the induction motor to a reduced supply at starting and then connecting it to the full voltage as the motor picks up sufficient speed. Fig: 3.24 shows the circuit arrangement for autotransformer starting. The tapping on the autotransformer is so set that when it is in the circuit, 65% to 80% of line voltage is applied to the motor.

At the instant of starting, the change-over switch is thrown to “start” position. This puts the autotransformer in the circuit and thus reduced voltage is applied to the circuit. Consequently, starting current is limited to safe value. When the motor attains about 80% of normal speed, the changeover switch is thrown to “run” position. This takes out the autotransformer from the circuit and puts the motor to full line voltage. Autotransformer starting has several advantages viz low power loss, low starting current and less radiated heat. For large machines (over 25 H.P.), this method of starting is often used. This method can be used for both star and delta connected motors.

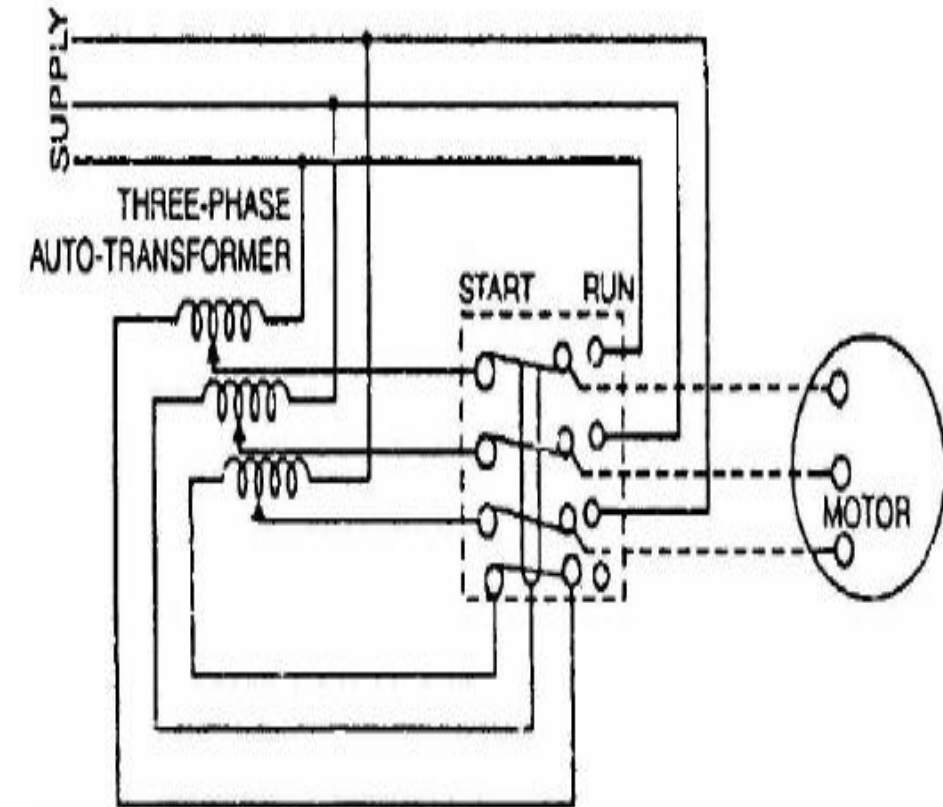


Fig: 3.24

**Relation between starting And F.L. torques.** Consider a star-connected squirrel-cage induction motor. If  $V$  is the line voltage, then voltage across motor phase on direct switching is  $V/\sqrt{3}$  and starting current is  $I_{st} = I_{sc}$ . In case of autotransformer, if a tapping of transformation ratio  $K$  (a fraction) is used, then phase voltage across motor is  $KV/\sqrt{3}$  and  $I_{st} = K I_{sc}$ ,

Now 
$$\frac{T_{st}}{T_f} = \left( \frac{I_{st}}{I_f} \right)^2 \times s_f = \left( \frac{K I_{sc}}{I_f} \right)^2 \times s_f = K^2 \left( \frac{I_{sc}}{I_f} \right)^2 \times s_f$$

$$\therefore \frac{T_{st}}{T_f} = K^2 \left( \frac{I_{sc}}{I_f} \right)^2 \times s_f$$

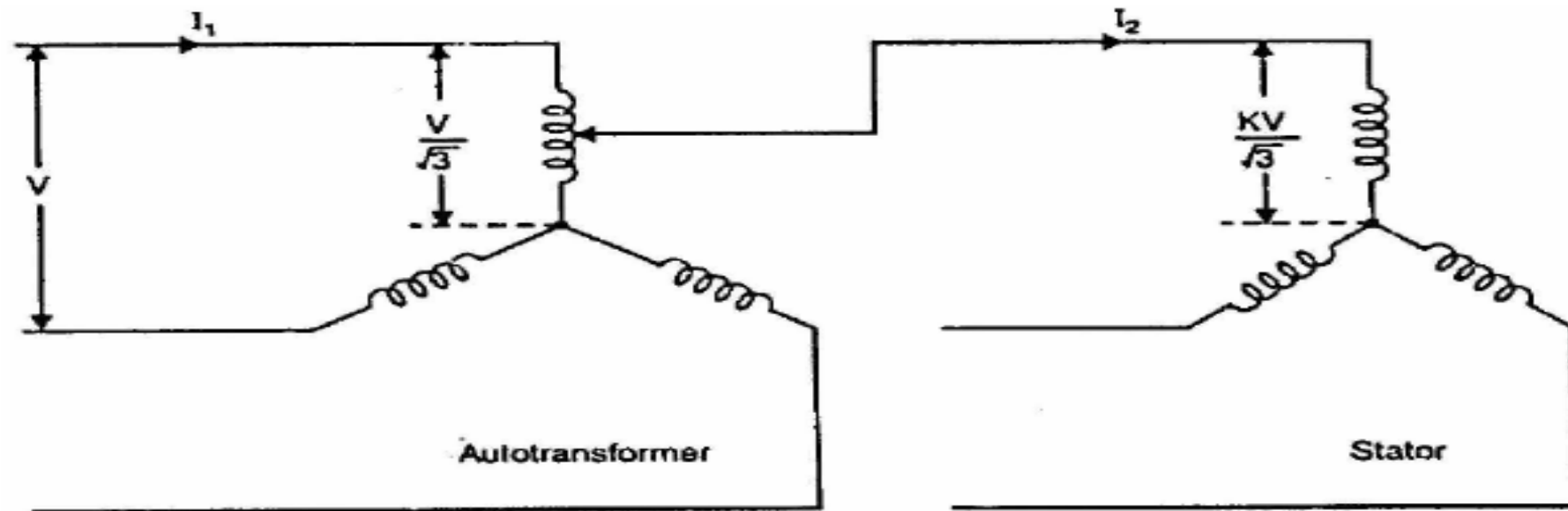


Fig: 3.25

The current taken from the supply or by autotransformer is  $I_1 = KI_2 = K^2 I_{sc}$ . Note that motor current is  $K$  times, the supply line current is  $K^2$  times and the starting torque is  $K^2$  times the value it would have been on direct-on-line starting.

(iv) *Star-delta starting*

The stator winding of the motor is designed for delta operation and is connected in star during the starting period. When the machine is up to speed, the connections are changed to delta. The circuit arrangement for star-delta starting is shown in Fig: 3.26.

The six leads of the stator windings are connected to the changeover switch as shown. At the instant of starting, the changeover switch is thrown to “Start” position which connects the stator windings in star. Therefore, each stator phase gets  $V/\sqrt{3}$  volts where  $V$  is the line voltage. This reduces the starting current. When the motor picks up speed, the changeover switch is thrown to “Run” position which connects the stator windings in delta. Now each stator phase gets full line voltage  $V$ . The disadvantages of this method are:

- (a) With star-connection during starting, stator phase voltage is  $1/\sqrt{3}$  times the line voltage. Consequently, starting torque is  $(1/\sqrt{3})^2$  or  $1/3$  times the value it would have with  $\Delta$ -connection. This is rather a large reduction in starting torque.
- (b) The reduction in voltage is fixed.

This method of starting is used for medium-size machines (upto about 25 H.P.).

**Relation between starting and F.L. torques.** In direct delta starting,

Starting current/phase,  $I_{sc} = V/Z_{sc}$  where  $V$  = line voltage

Starting line current =  $\sqrt{3} I_{sc}$

In star starting, we have,

Starting current/phase,  $I_{st} = \frac{V/\sqrt{3}}{Z_{sc}} = \frac{1}{\sqrt{3}} I_{sc}$

Now 
$$\frac{T_{st}}{T_f} = \left( \frac{I_{st}}{I_f} \right)^2 \times s_f = \left( \frac{I_{sc}}{\sqrt{3} \times I_f} \right)^2 \times s_f$$

or 
$$\frac{T_{st}}{T_f} = \frac{1}{3} \left( \frac{I_{sc}}{I_f} \right)^2 \times s_f$$

where  $I_{sc}$  = starting phase current (delta)  
 $I_f$  = F.L. phase current (delta)

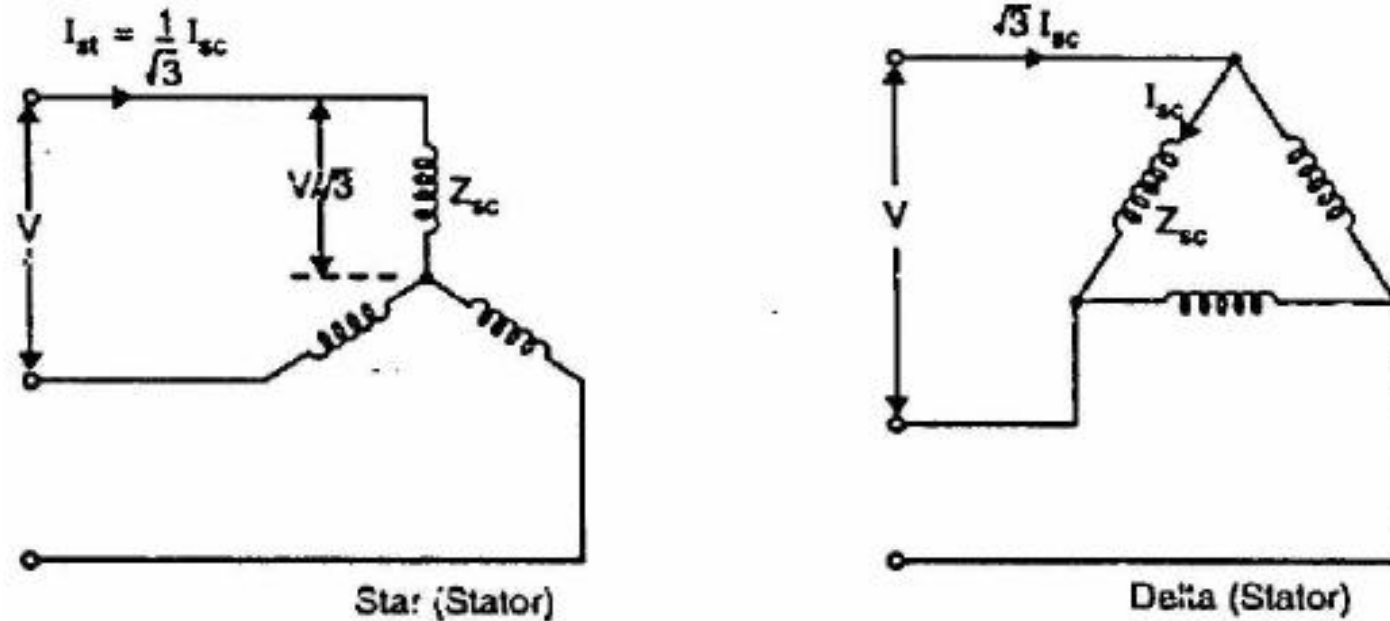


Fig: 3.26

Note that in star-delta starting, the starting line current is reduced to one-third as compared to starting with the winding delta connected. Further, starting torque is reduced to one-third of that obtainable by direct delta starting. This method is cheap but limited to applications where high starting torque is not necessary e.g., machine tools, pumps etc.

### 3.15.2 Starting of Slip-Ring Induction Motors

Slip-ring motors are invariably started by rotor resistance starting. In this method, a variable star-connected rheostat is connected in the rotor circuit through slip rings and full voltage is applied to the stator winding as shown in Fig: 3.27.

- (i) At starting, the handle of rheostat is set in the OFF position so that maximum resistance is placed in each phase of the rotor circuit. This reduces the starting current and at the same time starting torque is increased.
- (ii) As the motor picks up speed, the handle of rheostat is gradually moved in clockwise direction and cuts out the external resistance in each phase of the rotor circuit. When the motor attains normal speed, the change-over switch is in the ON position and the whole external resistance is cut out from the rotor circuit.

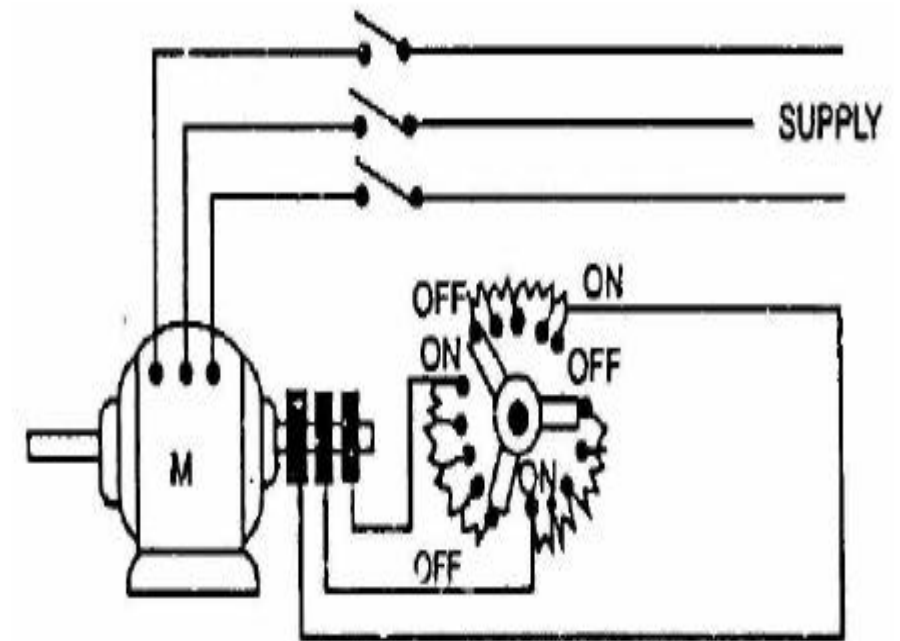


Fig: 3.27

### 3.16 Speed control of Three Phase Induction Motors

The induction machine, when operating from mains is essentially a constant speed machine. Many industrial drives, typically for fan or pump applications, have typically constant speed requirements and hence the induction machine is ideally suited for these. However, the induction machine, especially the squirrel cage type, is quite rugged and has a simple construction. Therefore it is good candidate for variable speed applications if it can be achieved.

#### 3.16.1 Speed control by changing applied voltage

From the torque equation of the induction machine we can see that the torque depends on the square of the applied voltage. The variation of speed torque curves with respect to the applied voltage is shown in Fig: 3.28. These curves show that the slip at maximum torque  $\hat{s}$  remains same, while the value of stall torque comes down with decrease in applied voltage. The speed range for stable operation remains the same.

Further, we also note that the starting torque is also lower at lower voltages. Thus, even if a given voltage level is sufficient for achieving the running torque, the machine may not start. This method of trying to control the speed is best suited for loads that require very little starting torque, but their torque requirement may increase with speed.

Fig: 3.28 also shows a load torque characteristic — one that is typical of a fan type of load. In a fan (blower) type of load, the variation of torque with speed is such that  $T \propto \omega^2$ .

Here one can see that it may be possible to run the motor to lower speeds within the range  $n_s$  to  $(1 - s)n_s$ . Further, since the load torque at zero speed is zero, the machine can start even at reduced voltages. This will not be possible with constant torque type of loads.

One may note that if the applied voltage is reduced, the voltage across the magnetising branch also comes down. This in turn means that the magnetizing current and hence flux level are reduced. Reduction in the flux level in the machine impairs torque production which is primarily the explanation for Fig: 3.28. If, however, the machine is running under lightly loaded conditions, then operating under rated flux levels is not required. Under such conditions, reduction in magnetizing current improves the power factor of operation. Some amount of energy saving may also be achieved.

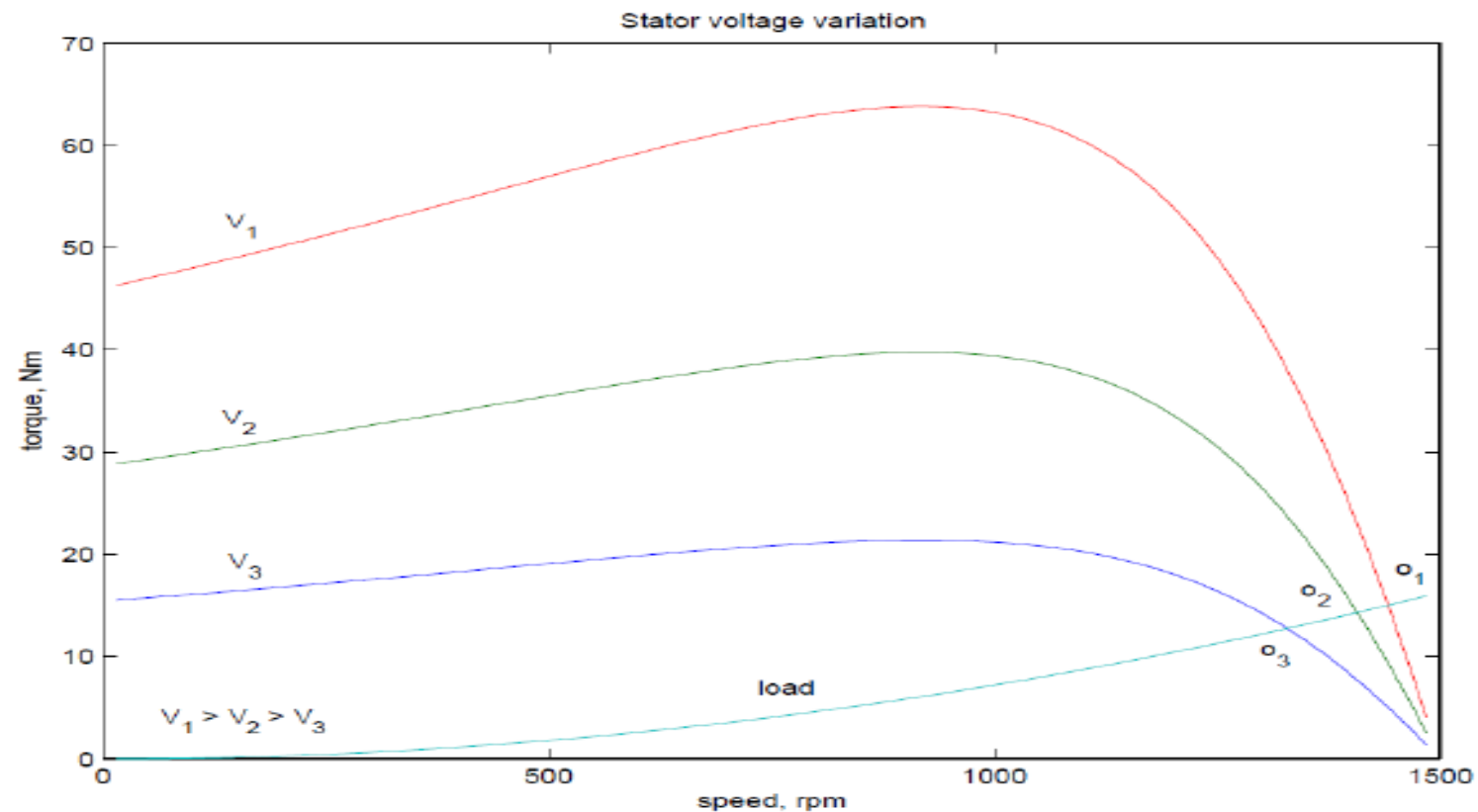


Fig: 3.28

Voltage control may be achieved by adding series resistors (a lossy, inefficient proposition), or a series inductor / autotransformer (a bulky solution) or a more modern solution using semiconductor devices. A typical solid state circuit used for this purpose is the AC voltage controller or AC chopper.

### 3.16.2 Rotor resistance control

The expression for the torque of the induction machine is dependent on the rotor resistance. Further the maximum value is independent of the rotor resistance. The slip at maximum torque is dependent on the rotor resistance. Therefore, we may expect that if the rotor resistance is changed, the maximum torque point shifts to higher slip values, while retaining a constant torque. Fig: 3.29 shows a family of torque-speed characteristic obtained by changing the rotor resistance.

Note that while the maximum torque and synchronous speed remain constant, the slip at which maximum torque occurs increases with increase in rotor resistance, and so does the starting torque. Whether the load is of constant torque type or fan-type, it is evident that the speed control range is more with this method. Further, rotor resistance control could also be used as a means of generating high starting torque.

For all its advantages, the scheme has two serious drawbacks. Firstly, in order to vary the rotor resistance, it is necessary to connect external variable resistors (winding resistance itself cannot be changed). This, therefore necessitates a slip-ring machine, since only in that case rotor terminals are available outside. For cage rotor machines, there are no rotor terminals. Secondly, the method is not very efficient since the additional resistance and operation at high slips entails dissipation.

The resistors connected to the slip-ring brushes should have good power dissipation capability. Water based rheostats may be used for this. A 'solid-state' alternative to a rheostat is a chopper controlled resistance where the duty ratio control of the chopper presents a variable resistance load to the rotor of the induction machine.

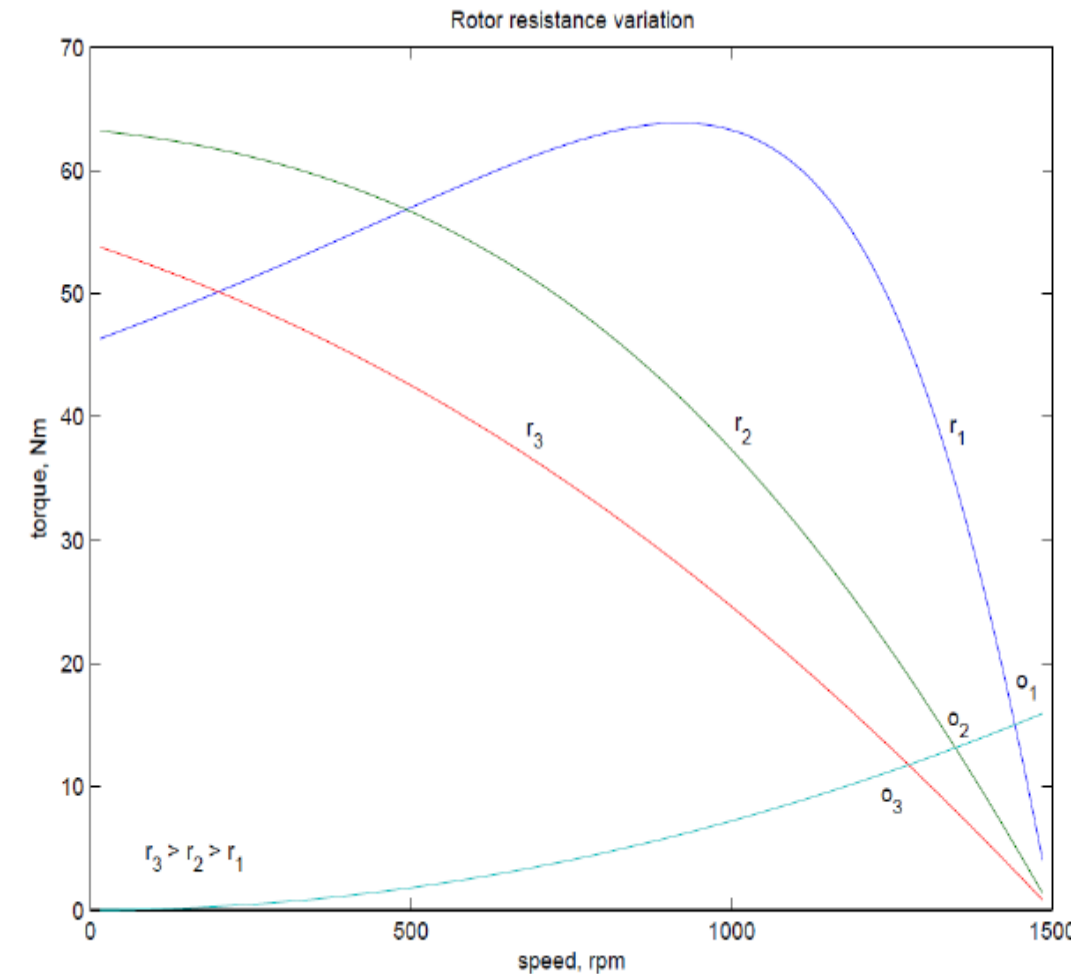


Fig: 3.29

### 3.16.3 Cascade control

The power drawn from the rotor terminals could be spent more usefully. Apart from using the heat generated in meaning full ways, the slip ring output could be connected to another induction machine. The stator of the second machine would carry slip frequency currents of

the first machine which would generate some useful mechanical power. A still better option would be to mechanically couple the shafts of the two machines together. This sort of a connection is called cascade connection and it gives some measure of speed control.

Let the frequency of supply given to the first machine be  $f_1$ , its number poles be  $p_1$ , and its slip of operation be  $s_1$ . Let  $f_2$ ,  $p_2$  and  $s_2$  be the corresponding quantities for the second machine. The frequency of currents flowing in the rotor of the first machine and hence in the stator of the second machine is  $s_1 f_1$ . Therefore  $f_2 = s_1 f_1$ . Since the machines are coupled at the shaft, the speed of the rotor is common for both. Hence, if  $n$  is the speed of the rotor in radians,

$$n = \frac{f_1}{p_1}(1 - s_1) = \pm \frac{s_1 f_1}{p_2}(1 - s_2).$$

Note that while giving the rotor output of the first machine to the stator of the second, the resultant stator mmf of the second machine may set up an air-gap flux which rotates in the same direction as that of the rotor, or opposes it. This results in values for speed as –

$$n = \frac{f_1}{p_1 + p_2} \quad \text{or} \quad n = \frac{f_1}{p_1 - p_2} \quad (s_2 \text{ negligible})$$

The latter expression is for the case where the second machine is connected in opposite phase sequence to the first. The cascade connected system can therefore run at two possible speeds.

Speed control through rotor terminals can be considered in a much more general way. Consider the induction machine equivalent circuit of Fig: 3.30, where the rotor circuit has been terminated with a voltage source  $E_r$ .

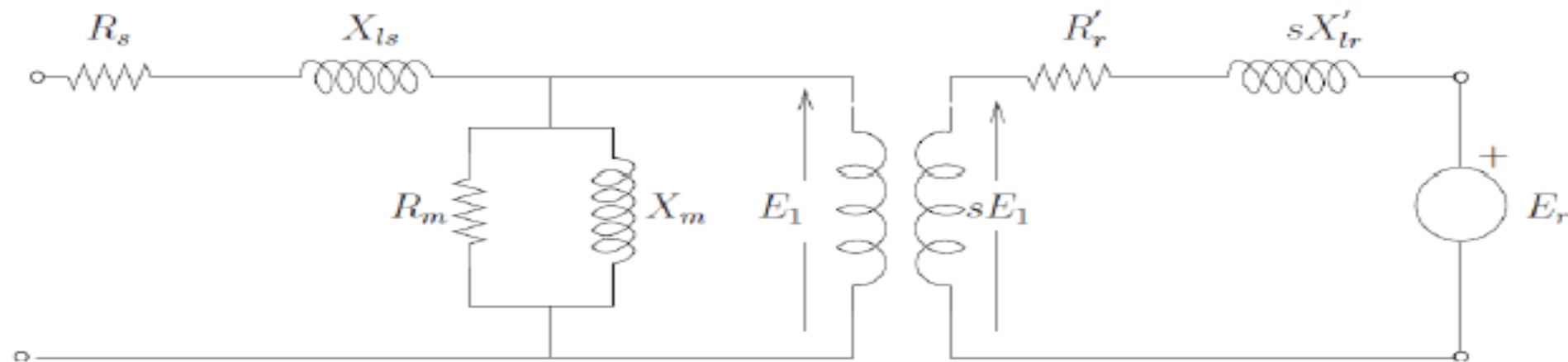


Fig: 3.30

If the rotor terminals are shorted, it behaves like a normal induction machine. This is equivalent to saying that across the rotor terminals a voltage source of zero magnitude is connected. Different situations could then be considered if this voltage source  $E_r$  had a non-zero magnitude. Let the power consumed by that source be  $P_r$ . Then considering the rotor side circuit power dissipation per phase

$$sE_1 I_2' \cos \phi_2 = I_2'^2 R_2 + P_r.$$

Clearly now, the value of  $s$  can be changed by the value of  $P_r$ . for  $P_r = 0$ , the machine is like a normal machine with a short circuited rotor. As  $P_r$  becomes positive, for all other circuit conditions remaining constant,  $s$  increases or in the other words, speed reduces. As  $P_r$  becomes negative, the right hand side of the equation and hence the slip decreases. The physical interpretation is that we now have an active source connected on the rotor side which is able to supply part of the rotor copper losses. When  $P_r = -I_2'^2 R_2$  the entire copper loss is supplied by the external source. The RHS and hence the slip is zero. This corresponds to operation at synchronous speed. In general the circuitry connected to the rotor may not be a simple resistor or a machine but a power electronic circuit which can process this power requirement. This circuit may drive a machine or recover power back to the mains. Such circuits are called static Kramer drives.

#### 3.16.4 Pole changing method

Sometimes induction machines have a special stator winding capable of being externally connected to form two different number of pole numbers. Since the synchronous speed of the induction machine is given by  $n_s = f_s/p$  (in rev. /s) where  $p$  is the number of pole pairs, this would correspond to changing the synchronous speed. With the slip now corresponding to the new synchronous speed, the operating speed is changed. This method of speed control is a stepped variation and generally restricted to two steps.

If the changes in stator winding connections are made so that the air gap flux remains constant, then at any winding connection, the same maximum torque is achievable. Such winding arrangements are therefore referred to as constant-torque connections. If however such connection changes result in air gap flux changes that are inversely proportional to the synchronous speeds, then such connections are called constant-horsepower type.

The following figure serves to illustrate the basic principle. Consider a magnetic pole structure consisting of four pole faces A, B, C, D as shown in Fig: 3.31.

Coils are wound on A & C in the directions shown. The two coils on A & C may be connected in series in two different ways — A2 may be connected to C1 or C2. A1 with the

Other terminal at C then form the terminals of the overall combination. Thus two connections result as shown in Fig: 3.32 (a) & (b).

Now, for a given direction of current flow at terminal A1, say into terminal A1, the flux directions within the poles are shown in the figures. In case (a), the flux lines are out of the pole A (seen from the rotor) for and into pole C, thus establishing a two-pole structure. In case (b) however, the flux lines are out of the poles in A & C. The flux lines will be then have to complete the circuit by flowing into the pole structures on the sides. If, when seen from the rotor, the pole emanating flux lines is considered as North Pole and the pole into which they enter is termed as south, then the pole configurations produced by these connections is a two-pole arrangement in Fig: 3.32(a) and a four-pole arrangement in Fig: 3.32 (b).

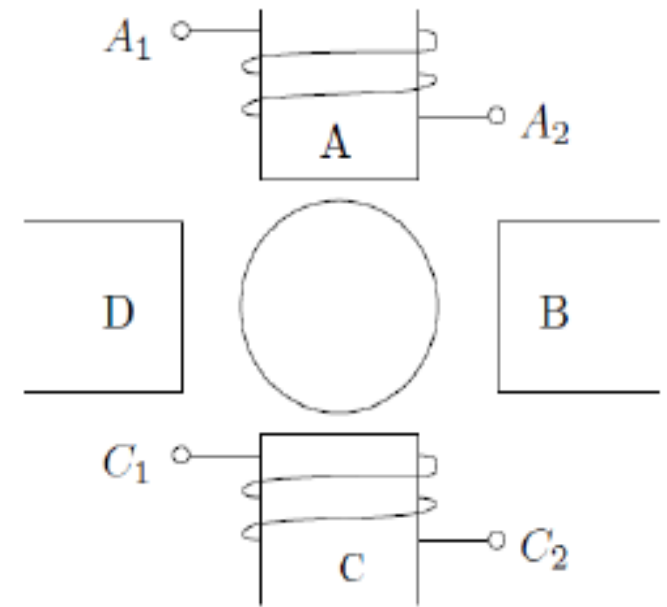


Fig: 3.31

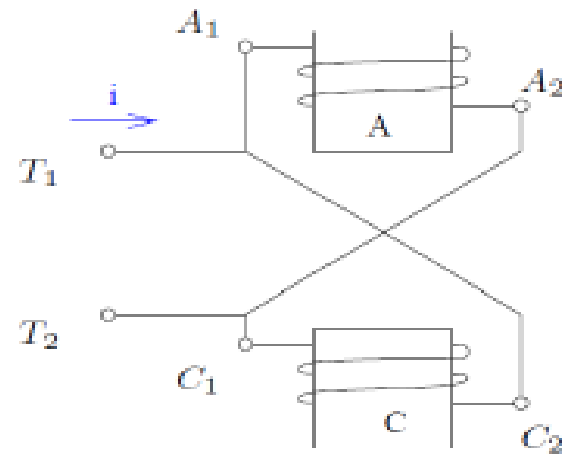
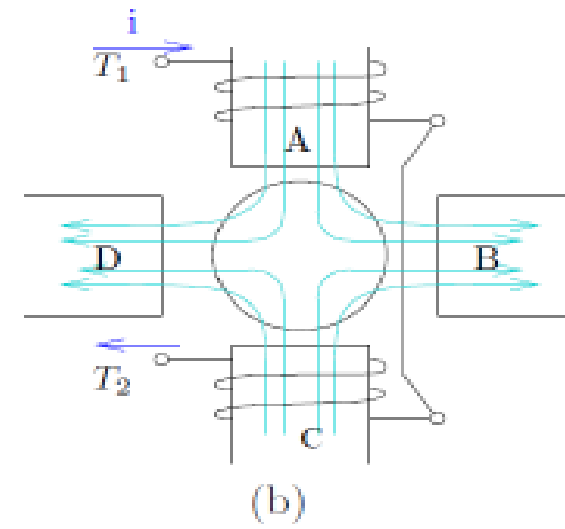
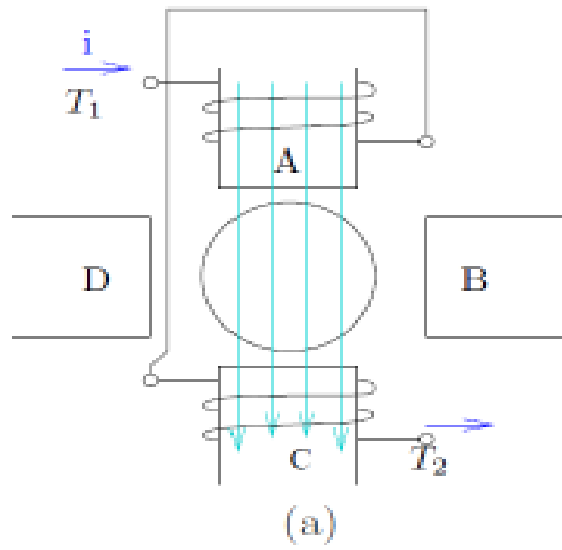


Fig: 3.32

Thus by changing the terminal connections we get either a two pole air-gap field or a four- pole field. In an induction machine this would correspond to a synchronous speed reduction in half from case (a) to case (b). Further note that irrespective of the connection, the applied voltage is balanced by the series addition of induced emf s in two coils. Therefore the air-gap flux in both cases is the same. Cases (a) and (b) therefore form a pair of constant torque connections.

Consider, on the other hand a connection as shown in the Fig: 3.32 (c). The terminals T1 and T2 are where the input excitation is given. Note that current direction in the coils now resembles that of case (b), and hence this would result in a four-pole structure. However, in Fig: 3.32 (c), there is only one coil induced emf to balance the applied voltage. Therefore flux in case (c) would therefore be halved compared to that of case (b) or case (a), for that matter). Cases (a) and (c) therefore form a pair of constant horse-power connections.

It is important to note that in generating a different pole numbers, the current through one coil (out of two, coil C in this case) is reversed. In the case of a three phase machine, the following example serves to explain this. Let the machine have coils connected as shown [C1 – C6] as shown in Fig: 3.33.

The current directions shown in  $C_1$  &  $C_2$  correspond to the case where  $T_1, T_2, T_3$  are supplied with three phase excitation and  $T_a, T_b$  &  $T_c$  are shorted to each other (STAR point). The applied voltage must be balanced by induced emf in one coil only ( $C_1$  &  $C_2$  are

parallel). If however the excitation is given to  $T_a, T_b$  &  $T_c$  with  $T_1, T_2, T_3$  open, then current through one of the coils ( $C_1$  &  $C_2$ ) would reverse. Thus the effective number of poles would increase, thereby bringing down the speed. The other coils also face similar conditions.

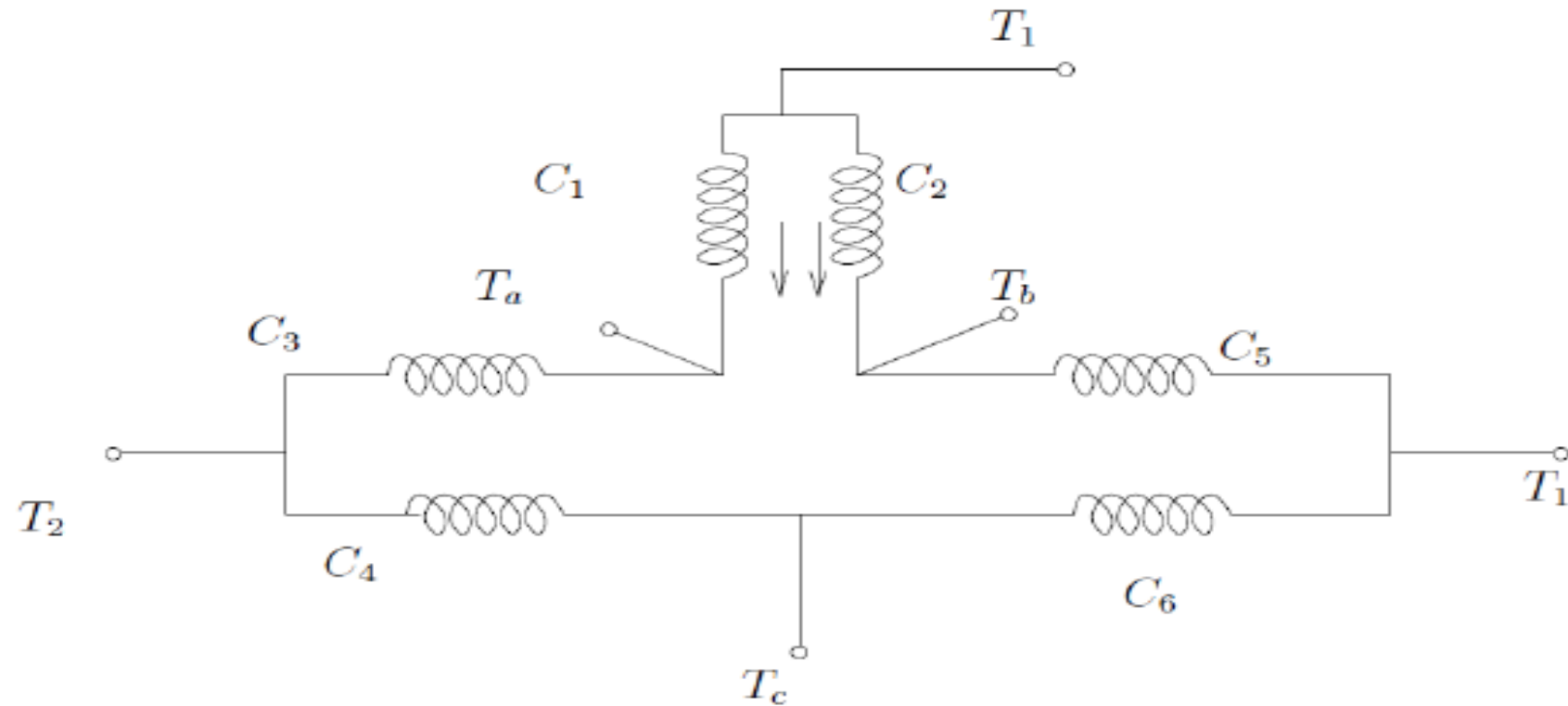


Fig: 3.33

### 3.16.5 Stator frequency control

The expression for the synchronous speed indicates that by changing the stator frequency also it can be changed. This can be achieved by using power electronic circuits called inverters which convert dc to ac of desired frequency. Depending on the type of control scheme of the inverter, the ac generated may be variable-frequency-fixed-amplitude or variable-frequency-variable-amplitude type. Power electronic control achieves smooth variation of voltage and frequency of the ac output. This when fed to the machine is capable of running at a controlled speed. However, consider the equation for the induced emf in the induction machine.

$$V = 4.44N\phi_m f$$

Where,  $N$  is the number of the turns per phase,  $\phi_m$  is the peak flux in the air gap and  $f$  is the frequency.

Actually, it is the voltage across the magnetizing branch of the exact equivalent circuit that must be maintained constant, for it is that which determines the induced emf. Under conditions where the stator voltage drop is negligible compared the applied voltage. In this mode of operation, the voltage across the magnetizing inductance in the 'exact' equivalent circuit reduces in amplitude with reduction in frequency and so does the inductive reactance. This implies that the current through the inductance and the flux in the machine remains constant. The speed torque characteristics at any frequency may be estimated as before. There is one curve for every excitation frequency considered corresponding to every

value of synchronous speed. The curves are shown below. It may be seen that the maximum torque remains constant.

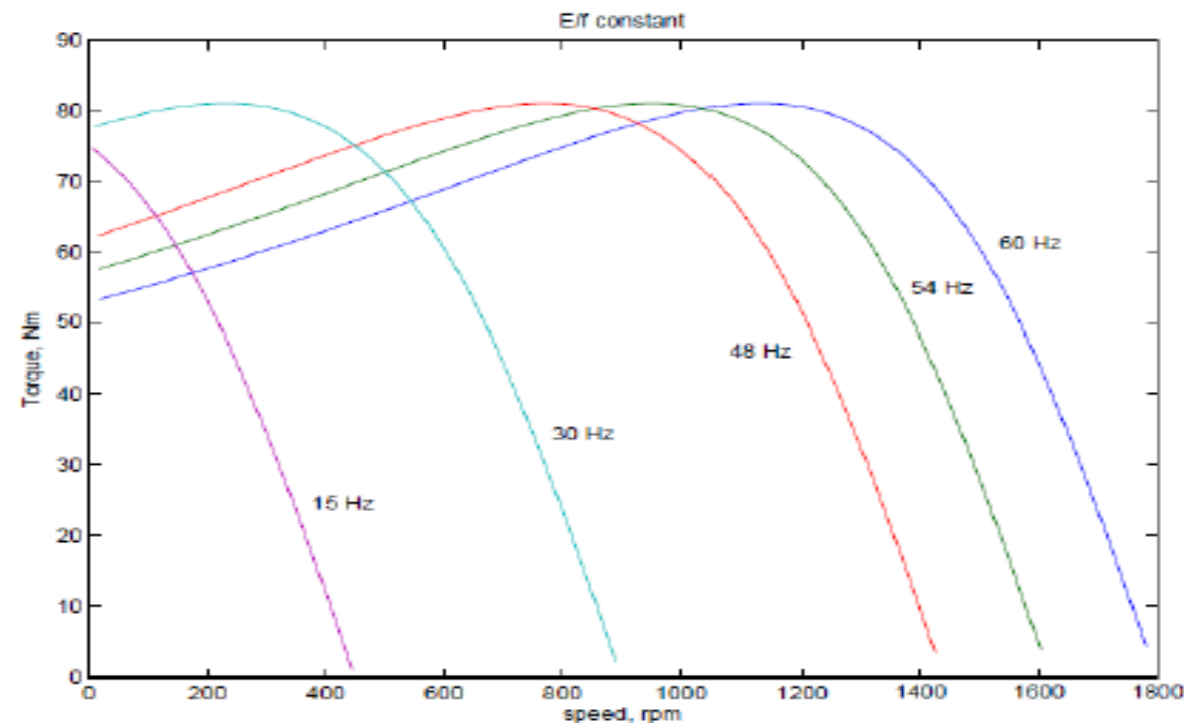


Fig: 3.34

### 3.17 Power Stages in an Induction Motor

The input electric power fed to the stator of the motor is converted into mechanical power at the shaft of the motor. The various losses during the energy conversion are:

#### *1. Fixed losses*

- (i) Stator iron loss
- (ii) Friction and windage loss

The rotor iron loss is negligible because the frequency of rotor currents under normal running condition is small.

#### *2. Variable losses*

- (i) Stator copper loss
- (ii) Rotor copper loss

Fig: 3.37 shows how electric power fed to the stator of an induction motor suffers losses and finally converted into mechanical power.

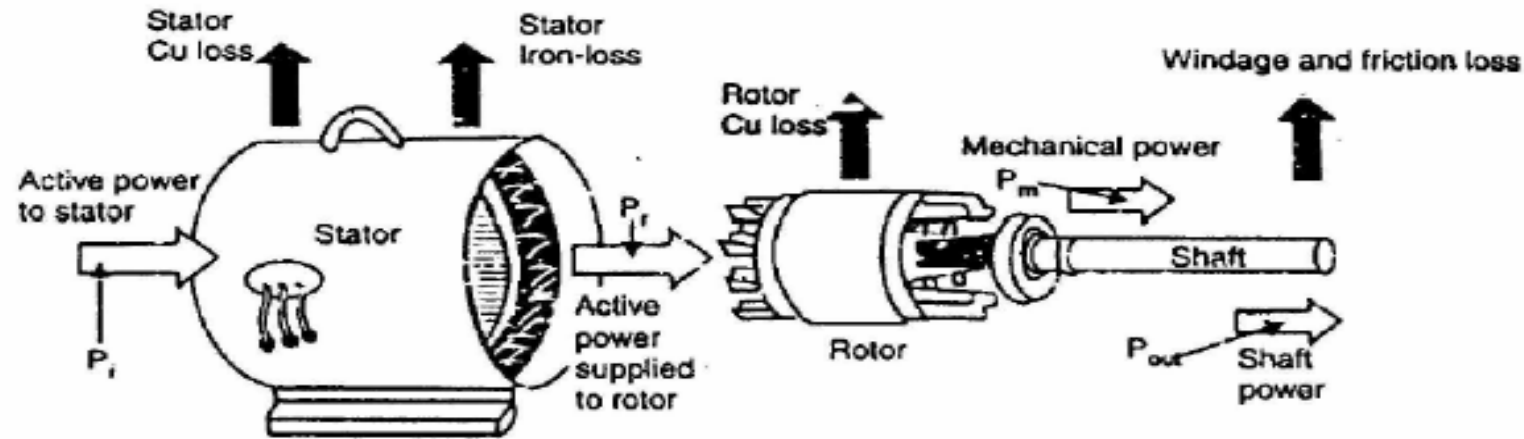


Fig: 3.37

The following points may be noted from the above diagram:

- (i) Stator input,  $P_i$  = Stator output + Stator losses  
= Stator output + Stator Iron loss + Stator Cu loss
- (ii) Rotor input,  $P_r$  = Stator output

(iii) Mechanical power available,  $P_m = P_r - \text{Rotor Cu loss}$

This mechanical power available is the gross rotor output and will produce a gross torque  $T_g$ .

(iv) Mechanical power at shaft,  $P_{out} = P_m - \text{Friction and windage loss}$

Mechanical power available at the shaft produces a shaft torque  $T_{sh}$ .

Clearly,  $P_m - P_{out} = \text{Friction and windage loss}$ .

### 3.18 Double Cage Induction Motor

One of the advantages of the slip-ring motor is that resistance may be inserted in the rotor circuit to obtain high starting torque (at low starting current) and then cut out to obtain optimum running conditions. However, such a procedure cannot be adopted for a squirrel cage motor because its cage is permanently short-circuited. In order to provide high starting torque at low starting current, double-cage construction is used.

### Construction

As the name suggests, the rotor of this motor has two squirrel-cage windings located one above the other as shown in Fig: 3.38(i).

*The outer winding* consists of bars of smaller cross-section short-circuited by end rings. Therefore, the resistance of this winding is high. Since the outer winding has relatively open slots and a poorer flux path around its bars [See Fig: 3.38(ii)], it has a low inductance. Thus the resistance of the outer squirrel-cage winding is high and its inductance is low.

*The inner winding* consists of bars of greater cross-section short-circuited by end rings. Therefore, the resistance of this winding is low. Since the bars of the inner winding are thoroughly buried in iron, it has a high inductance [See Fig: 3.38(ii)]. Thus the resistance of the inner squirrel cage winding is low and its inductance is high.

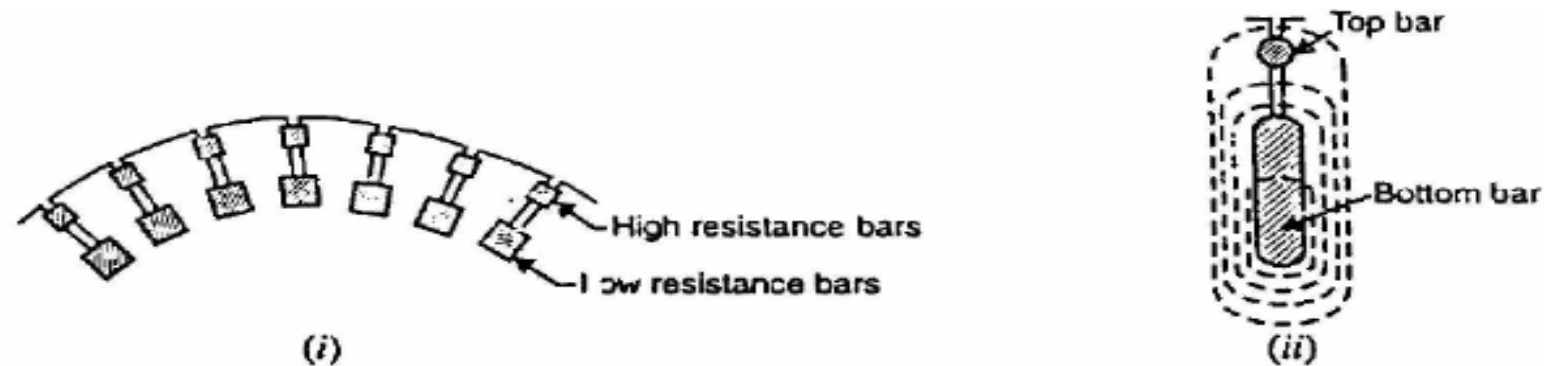


Fig: 3.38

### *Working*

When a rotating magnetic field sweeps across the two windings, equal e.m.f.s are induced in each.

(i) At starting, the rotor frequency is the same as that of the line (i.e., 50 Hz), making the reactance of the lower winding much higher than that of the upper winding. Because of the high reactance of the lower winding, nearly all the rotor current flows in the high-resistance outer cage winding. This provides the good starting characteristics of a high-resistance cage winding. Thus the outer winding gives high starting torque at low starting current.

(ii) As the motor accelerates, the rotor frequency decreases, thereby lowering the reactance of the inner winding, allowing it to carry a larger proportion of the total rotor current. At the normal operating speed of the motor, the rotor frequency is so low (2 to 3 Hz) that nearly all the rotor current flows in the low-resistance inner cage winding. This results in good operating efficiency and speed regulation.

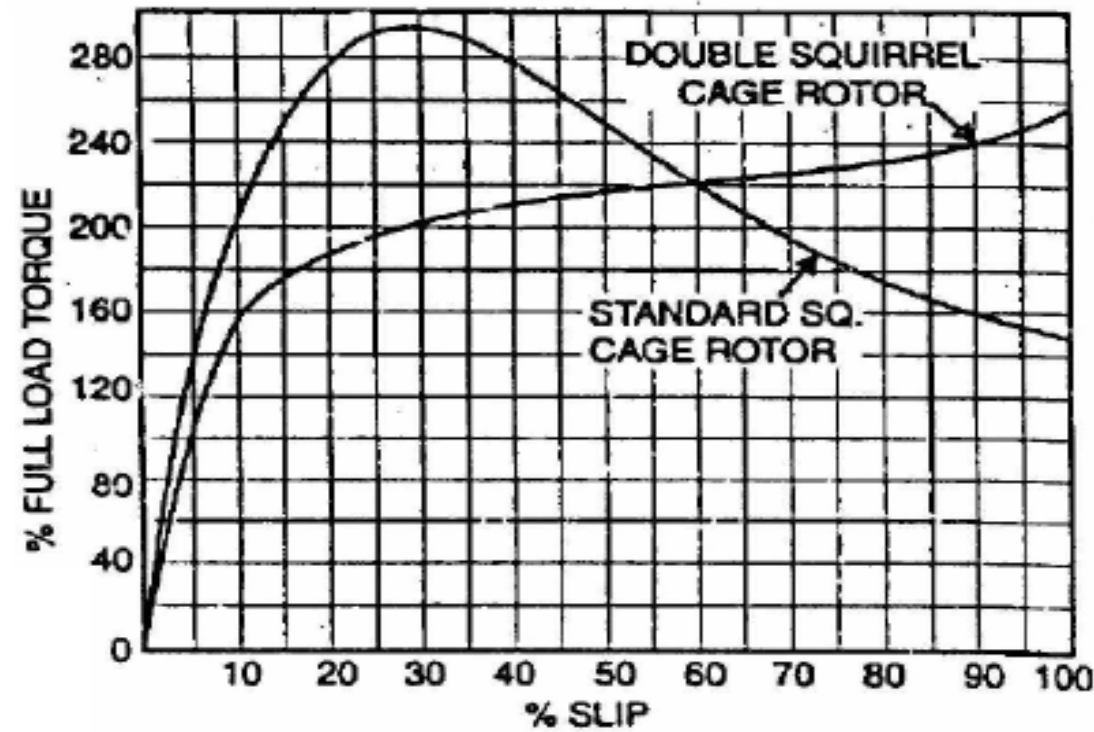


Fig: 3.39

Fig: 3.39 shows the operating characteristics of double squirrel-cage motor. The starting torque of this motor ranges from 200 to 250 percent of full-load torque with a starting current of 4 to 6 times the full-load value. It is classed as a high-torque, low starting current motor.

### 3.19 Cogging and Crawling of Induction Motor

#### *Crawling of induction motor*

Sometimes, squirrel cage induction motors exhibit a tendency to run at very slow speeds (as low as one-seventh of their synchronous speed). This phenomenon is called as crawling of an induction motor.

This action is due to the fact that, flux wave produced by a stator winding is not purely sine wave. Instead, it is a complex wave consisting of a fundamental wave and odd harmonics like 3rd, 5th, 7th etc. The fundamental wave revolves synchronously at synchronous speed  $N_s$  whereas 3rd, 5th, 7th harmonics may rotate in forward or backward direction at  $N_s/3$ ,  $N_s/5$ ,  $N_s/7$  speeds respectively. Hence, harmonic torques are also developed in addition with fundamental torque.

3rd harmonics are absent in a balanced 3-phase system. Hence 3rd harmonics do not produce rotating field and torque. The total motor torque now consists of three components as: (i) the fundamental torque with synchronous speed  $N_s$ , (ii) 5th harmonic torque with synchronous speed

$N_s/5$ , (iv) 7th harmonic torque with synchronous speed  $N_s/7$  (provided that higher harmonics are neglected).

Now, 5th harmonic currents will have phase difference of

$$5 \times 120 = 600^\circ = 2 \times 360 - 120 = -120^\circ.$$

Hence the revolving speed set up will be in reverse direction with speed  $N_s/5$ . The small amount of 5th harmonic torque produces breaking action and can be neglected.

The 7th harmonic currents will have phase difference of

$$7 \times 120 = 840^\circ = 2 \times 360 + 120 = +120^\circ.$$

Hence they will set up rotating field in forward direction with synchronous speed equal to  $N_s/7$ . If we neglect all the higher harmonics, the resultant torque will be equal to sum of fundamental torque and 7th harmonic torque. 7<sup>th</sup> harmonic torque reaches its maximum positive value just before  $1/7^{\text{th}}$  of  $N_s$ . If the mechanical load on the shaft involves constant load torque, the torque developed by the motor may fall below this load torque. In this case, motor will not accelerate up to its normal speed, but it will run at a speed which is nearly  $1/7^{\text{th}}$  of its normal speed as shown in Fig: 3.40. *This phenomenon is called as crawling of induction motors.*

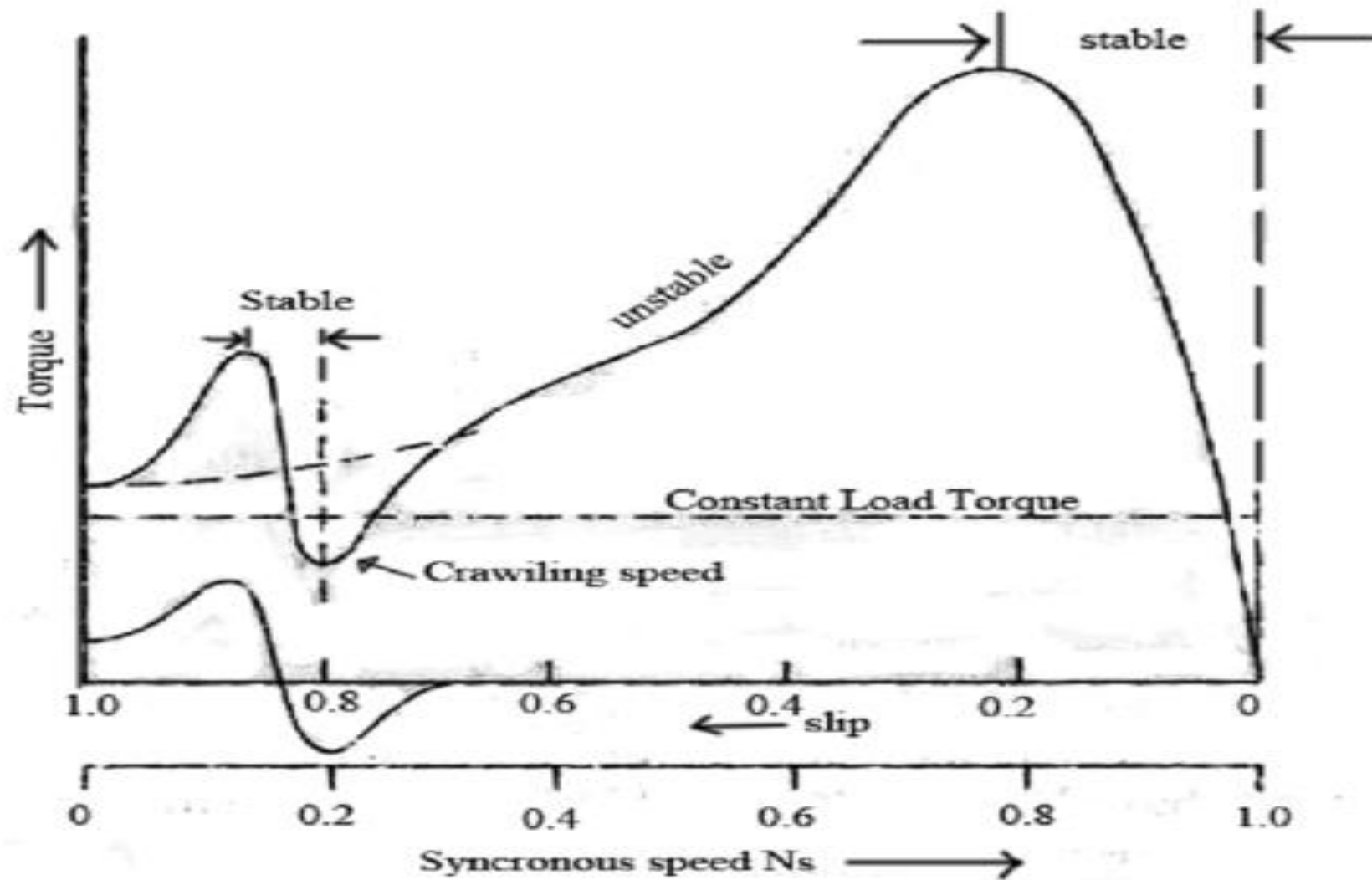


Fig: 3.40

### *Cogging (Magnetic Locking or Teeth Locking) of induction motor*

Sometimes, the rotor of a squirrel cage induction motor refuses to start at all, particularly if the supply voltage is low. This happens especially when number of rotor teeth is equal to number of stator teeth, because of magnetic locking between the stator teeth and the rotor teeth. When the rotor teeth and stator teeth face each other, the reluctance of the magnetic path is minimum that is why the rotor tends to remain fixed. This phenomenon is called cogging or magnetic locking of induction motor.

### **3.20 Induction Generator**

When a squirrel cage induction motor is energized from a three phase power system and is mechanically driven above its synchronous speed it will deliver power to the system. An induction generator receives its excitation (magnetizing current) from the system to which it is connected. It consumes rather than supplies reactive power (KVAR) and supplies only real power (KW) to the system. The KVAR required by the induction generator plus the KVAR requirements of all other loads on the system must be supplied from synchronous generators or static capacitors on the system.

Operating as a generator at a given percentage slip above synchronous speed, the torque, current, efficiency and power factor will not differ greatly from that when operating as a motor. The same slip below synchronous speed, the shaft torque and electric power flow is reversed. Typical speed torque characteristic of induction generator is shown in Fig: 3.41.

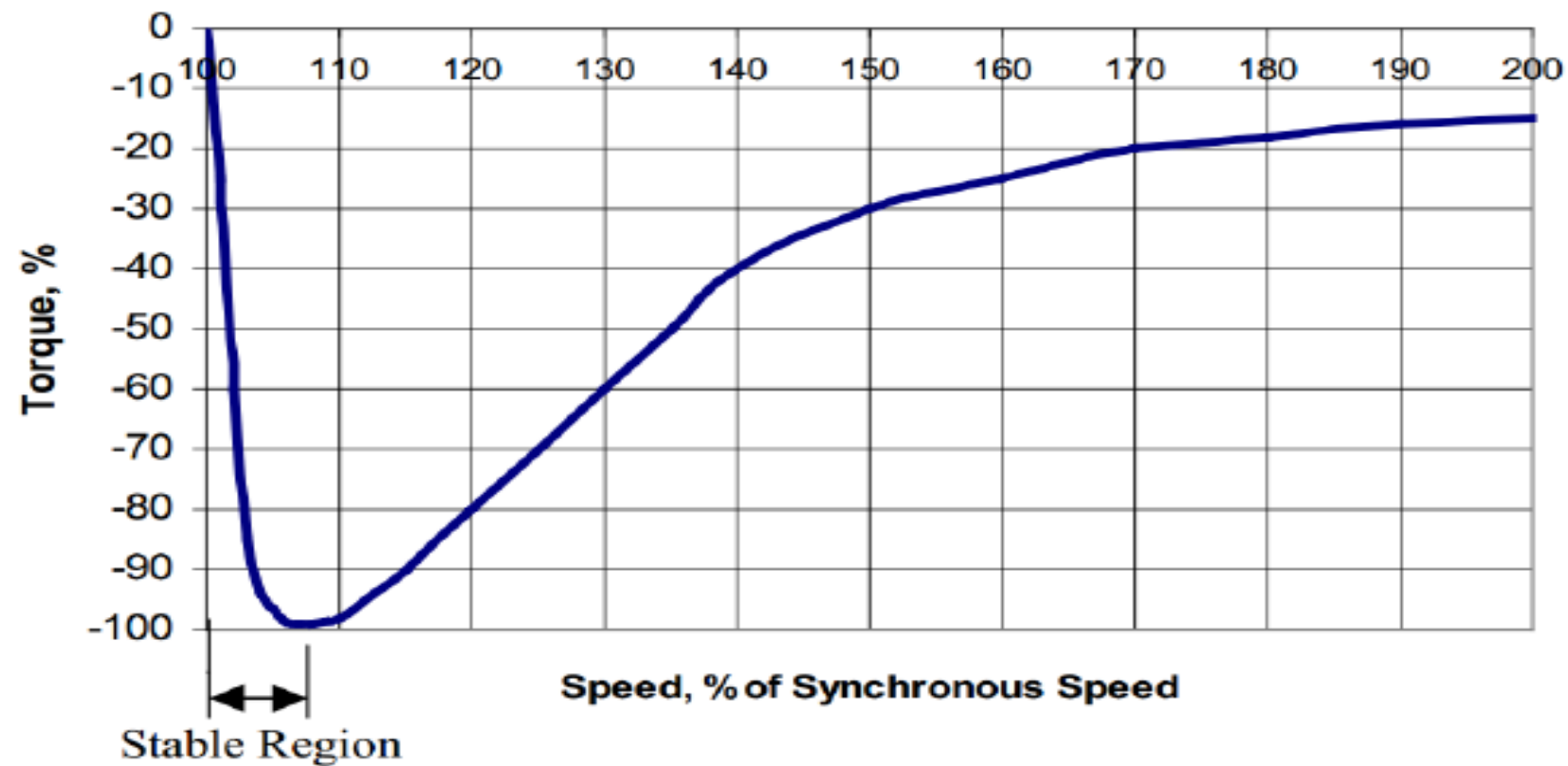


Fig: 3.41

Now for example, a 3600 RPM squirrel cage induction motor which delivers full load output at 3550 RPM as a motor will deliver full rated power as a generator at 3650 RPM. If the half-load motor speed is 3570 RPM, the output as a generator will be one-half of rated value when driven at 3630 RPM, etc. Since the induction generator is actually an induction motor being driven by a prime mover, it has several advantages.

1. It is less expensive and more readily available than a synchronous generator.
2. It does not require a DC field excitation voltage.
3. It automatically synchronizes with the power system, so its controls are simpler and less expensive.

The principal disadvantages of an induction generator are listed below

1. It is not suitable for separate, isolated operation
2. It consumes rather than supplies magnetizing KVAR
3. It cannot contribute to the maintenance of system voltage levels (this is left entirely to the synchronous generators or capacitors)
4. In general it has a lower efficiency.

### *Induction Generator Application*

As energy costs so high, energy recovery became an important part of the economics of most industrial processes. The induction generator is ideal for such applications because it requires very little in the way of control system or maintenance.

Because of their simplicity and small size per kilowatt of output power, induction generators are also favoured very strongly for small windmills. Many commercial windmills are designed to operate in parallel with large power systems, supplying a fraction of the customer's total power needs. In such operation, the power system can be relied on for voltage & frequency control, and static capacitors can be used for power-factor correction.